

On the Concrete Security of BBS/BBS+ Signatures

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Abstract. BBS/BBS+ signatures are the most promising solution to instantiate practical and light-weight anonymous credentials. They underlie standardization efforts by the W3C and the IRTF. Due to their potential for large scale deployment, it is paramount to understand their concrete security, but a number of questions have been left open by prior works. To this end, the security proofs by Au et al. (SCN '06), Camenisch et al. (TRUST '16), and Tessaro and Zhu (EUROCRYPT '23) show reductions from q -SDH in groups of prime order p , where q is the number of issued signatures.

However, these prior works left the possibility open that BBS/BBS+ is *even more secure* than what can be guaranteed by such proofs. Indeed, while the q -SDH assumption is subject to an attack that uses $O(\sqrt{p/q})$ group exponentiations (Cheon, EUROCRYPT '06) for several choices of q , no attack with a similar complexity appears to affect either of BBS+ and *deterministic* BBS, for which the best known attacks amount to recovering the secret key by breaking the discrete logarithm problem. The assumption that this attack is best possible also seemingly justifies the choice of parameters in practice.

Our result shows that this expectation is not true. We show new attacks against BBS+ and *deterministic* BBS which, after seeing q signatures, allow us to recover the secret key with the same complexity as solving the $\Theta(q)$ -Discrete Logarithm problem, which in turn is proportional to $O(\sqrt{p/q})$ for many choices of q . Further, we also extend the attack to a reduction showing that the security of BBS+ and deterministic BBS implies the $\Theta(q)$ -SDH assumption.

1 Introduction

A common approach to constructing many privacy-preserving primitives is to rely on special signature schemes with algebraic structures allowing them to support efficient zero-knowledge proofs of knowledge of valid message-signature pairs, and allowing them to be seen as signing an algebraic commitment of the message. Camenisch and Lysyanskaya [CL03, CL04] refer to these as “signatures with efficient protocols.”

BBS signatures [BBS04, ASMC13, CDL16, TZ23] are the to-date most popular such scheme, due to their efficiency and simplicity. They have been used as a building block for anonymous credentials [CL04, TAKS07], direct anonymous attestation [Che10, BL10, CDL16], and k -time anonymous authentication [ASMC13]. Recently, there have been renewed interests with industry implementations [mat, mic], standardization efforts by the W3C Verifiable Credentials working group [w3c], and by the Decentralized Identity Foundation, the latter leading to an RFC draft by the IRTF [LKWL25].

In view of their potential large-scale deployment, it is very important to understand the concrete security of BBS, which in turn affects its parameter choice. In this context, a number of questions have remained open, and the main goal of this paper is to close these gaps with new cryptanalysis.

BBS SIGNATURE AND ITS HISTORY. The BBS signature scheme was first stated in [CL04] without a security proof, taking inspiration from ideas from the concurrent work by Boneh, Boyen, and Shacham [BBS04], and hence the name. (Although the well-known BBS *group* signature is actually a distinct scheme.) Defined over pairing-friendly groups, the scheme signs a message $\mathbf{m} \in \mathbb{Z}_p^\ell$ into a signature $\sigma = (A, e)$ consisting of a group element $A \in \mathbb{G}_1$ and a scalar $e \in \mathbb{Z}_p$ such that

$$A = \frac{1}{x + e} \left(G_1 + \sum_{i=1}^{\ell} \mathbf{m}[i] \mathbf{H}[i] \right),$$

where x is the secret key and $G_1, H[1], \dots, H[\ell]$ are generators of $\mathbb{G}_1$. The verification can be done using the public key $X_2 = xG_2 \in \mathbb{G}_2$ and the bilinear pairing. Au, Susilo, and Mu [ASMC13] later gave a provably secure and slightly less efficient version, called BBS+¹, with signatures including an additional scalar. Their security is based on the q -SDH assumption² in type-2 pairings. After that, Camenisch, Drijvers, and Lehmann [CDL16] adapted the proof to the more efficient type-3 pairings.

Recently, Tessaro and Zhu [TZ23] revisited the original BBS signature and showed its strong unforgeability (SUF) based on the q -SDH assumption in the standard model. Additionally, they gave a tight security proof in the Algebraic Group Model (AGM) [FKL18] based on the q -DL assumption. Following after their work, the RFC draft [LKWL25] transitioned to cover BBS signatures instead of BBS+.

REGARDING CONCRETE PARAMETERS. The RFC draft suggested an implementation using the elliptic curve BLS12-381 [Bow17], which has roughly 256-bit prime order. However, the q -SDH/ q -DL problem, which the security of BBS relies on, admits Cheon’s attack [Che06] (and also a similar attack by Brown and Gallant [BG04]), which uses $O(\sqrt{p/q})$ group exponentiations for certain (but rather flexible) choices of q . As a consequence, the security of q -SDH problem on BLS12-381 curve is roughly 110-bit for $q = 2^{36}$.

In their recent work, Tessaro and Zhu [TZ23] speculated that *the actual security level of BBS might not be affected by Cheon’s attack*. Their reasoning is that the only known concrete attack against BBS is by the work of Jao and Yoshida’s [JY09], achieving the same complexity as Cheon’s attack, but their attack hinges on receiving different signatures on the *same message*. This situation, however, does not occur with the *deterministic* version of BBS used in practice (see e.g., the specification in [LKWL25, Section 3.6.1.]), or even in general, we cannot expect many signatures on the same message. Similarly, we do not know of such attack even for randomized BBS+, as one can effectively view BBS+ signatures as BBS signatures of random messages, which are unlikely to be equal. Following this reasoning, the implementations of BBS and BBS+ using the suggested BLS12-381 curve should still provide roughly 128-bit security.

OUR CONTRIBUTIONS. Our result will prove that the above speculation is actually *false*. Indeed, we give new attacks against deterministic BBS, as well as randomized BBS+, using $O(\sqrt{p/q})$ exponentiations matching the complexity of Jao and Yoshida’s attack and consequently the lower-bound in the Generic Group Model [Sho97, Mau05]. This confirms that in order to achieve 128-bit security, the implementations of BBS and BBS+ should use curves of prime order larger than 2^{256} . We note that our attacks also extend to derandomized BBS+. Our attacks are formulated as a reduction from security of (deterministic) BBS and BBS+ signatures to the hardness of $\Theta(q)$ -DL assumption (where one can apply Cheon’s attack). We also extend such reduction to the hardness of the $\Theta(q)$ -SDH assumption, which surprisingly requires a more complex analysis.

THEORETICAL CONSEQUENCES. Our attacks can be viewed as reductions showing the hardness of $\Theta(q)$ -DL (and $\Theta(q)$ -SDH) assumption from the security of (deterministic) BBS and BBS+ signatures against adversaries making q queries. Combined with a recent work by Lu and Zhandry [LZ24], which separates group-based q -type assumptions from static assumptions, this implies that the security of (deterministic) BBS and BBS+ cannot be based on any static assumptions, even with non-tight reductions.

PRACTICAL CONSEQUENCES. Because non-tight bounds are more common in public-key cryptography than, e.g., in symmetric cryptography, it has been common to adopt a cryptanalytic view for setting parameters. Here, our attacks with $O(\sqrt{p/q})$ complexity now put all schemes (BBS and BBS+, deterministic or randomized) at the same security level, and BBS gives efficiency advantages that are already utilized by industry implementations and standards.

Still, if concrete security is of concern, we do caution the choice of BLS12-381 (or similar curves with group size of roughly 256 bits) as the elliptic curve of choice whenever 128-bit of security are needed and/or it is hard to practically bound q . One may allow for multiple choice of curves, including those with approximate group size of 384 bits, as done e.g. in the recently finalized VOPRF RFC [DFHSW23].

¹ Note that the same signature scheme has also been called SDH-based CL signatures (see e.g., [Sch15]), referring to the scheme stated by Camenisch and Lysyanskaya [CL04].

² The value q denotes the number of signing queries.

RELATED WORKS. In pairing-free groups, BBS/BBS+ signatures have also been formalized as algebraic message authentication codes (algebraic MACs) (see e.g., as in [BBDT16, Orr24]), where verification is done via the secret key instead of publicly using pairings. Algebraic MACs are important building blocks for constructing keyed-verification anonymous credentials [CMZ14]. A recent candidate for the EU Digital Identity Wallet [ARF24, Ger24], called BBS# [TD, DDKT25], is also constructed on top of this BBS-based MAC. Notably, our attacks do not rely on the existence of bilinear pairings (nor the type of pairing groups); hence, they also apply to these BBS-based schemes and should affect their choice of parameters/elliptic curves.

The attack of Jao and Yoshida also has applications in contexts other than signatures. In particular, Lauer [Lau18] has adapted the attack to show equivalence of the q -Bilinear Decisional Diffie-Hellman assumption and the security of Dodis-Yampolskiy [DY05] and Boneh-Montgomery-Raghuathan [BMR10] verifiable random functions. Importantly, we note that their work does not overlap with our results.

PAPER OUTLINE. We introduce basic preliminaries in Section 2 and the specifications of BBS in Section 3. In Section 4, we give our attack on deterministic BBS as a reduction from SUF security to $\Theta(q)$ -DL assumption and discuss minor extensions. In Section 5, we extend the former attack into a reduction to $\Theta(q)$ -SDH assumption. We discuss further implications of our attacks and reductions in Section 6.

1.1 Technical Overview

Our starting point is Jao and Yoshida’s work [JY09] which showed that the security of Boneh-Boyen signatures [BB08] is equivalent to the q -SDH assumption. Their work implies an attack on Boneh-Boyen signatures that requires $O(\sqrt{p/q})$ group exponentiations. It was observed in [TZ23] that the attack applies to *randomized* BBS signatures as well, but only when all the messages signed are identical. Towards an attack on the deterministic BBS and randomized BBS+ signatures, we made a key observation, which we detail below, that lets us apply similar techniques to forge a signature even when given signatures on distinct messages.

RECALLING JAO AND YOSHIDA’S ATTACK. We first recall how the attack of Jao and Yoshida can be applied to BBS signatures, and for simplicity, we only consider message length $\ell = 1$. The attack first issues q signing queries on the same message $m = 0 \in \mathbb{Z}_p$, which results in q signatures of the form (A_i, e_i) such that

$$A_i = \frac{1}{x + e_i} G_1 ,$$

where x is the secret key and G_1 is a generator of $\mathbb{G}_1$. Assuming that $e_1, \dots, e_q$ are all distinct (which occurs with overwhelming probability), the attack linearly combines the A_i ’s into group elements of the form $x^k \bar{G}_1$ for all $k \in [0..q-1]$ for some group element $\bar{G}_1$, particularly of the form $p(x)^{-1} G_1$ with $p(X) = \prod_{i=1}^q (X + e_i) \in \mathbb{Z}_p[X]$ be a degree q polynomial.

To do this, we first denote $p_i(X) = \prod_{j \neq i} (X + e_j)$ for all $i \in [q]$. Then, we recall that any polynomial $g \in \mathbb{Z}_p[X]$ of degree less than q can be uniquely represented as a linear combination of the Lagrange basis polynomials denoted $\{L_i(X) = p_i(X)/p_i(-e_i)\}_{i \in [q]}$. In particular, we can write $g(X) = \sum_{i=1}^q g(-e_i) L_i(X)$. Hence, for any such polynomial g ,

$$\sum_{i=1}^q \frac{g(-e_i)}{p_i(-e_i)} A_i = \sum_{i=1}^q \frac{g(-e_i)}{p_i(-e_i)(x + e_i)} G_1 = \frac{1}{p(x)} \sum_{i=1}^q g(-e_i) \frac{p_i(x)}{p_i(-e_i)} G_1 = \frac{g(x)}{p(x)} G_1 .$$

Note in particular that the second equality follows from $X + e_i = p(X)/p_i(X)$, and the third equality follows from the fact about Lagrange basis polynomials. Then, picking $g(X) = X^k$ for $k \in [0..q-1]$ gives us a $(q-1)$ -DL instance. Finally, we can apply Cheon’s attack [Che06] (or a similar attack by Brown and Gallant [BG04]), which solves a d -DL instance in $O(\sqrt{p/d})$ group exponentiations for a divisor d of $p-1$. For this attack, we accordingly set $q = d + 1$.

OUR ATTACK ON DETERMINISTIC BBS. For deterministic BBS, since the signing queries on the same message will always result in the same signature, the same attack does not apply. Thus, our new attack needs to make queries on q distinct messages $m_1, \dots, m_q \in \mathbb{Z}_p$, resulting in signatures (A_i, e_i) where

$$A_i = \frac{G_1 + m_i H}{x + e_i}.$$

In order to apply Cheon's attack, the goal is to compute group elements of the form $x^k \overline{G}_1$ for some generator $\overline{G}_1$ from linear combinations of A_i . However, the approach by Jao and Yoshida does not quite work due to the introduction of the $m_i H$ term in the representation of A_i .

Our key observation is that any linear combination of A_i 's would still yield a group element of the form $p(x)^{-1}(g_0(x)G_1 + g_1(x)H)$ for some polynomials $g_0, g_1 \in \mathbb{Z}_p[X]$ of degree less than q . To see this, consider the linear combination with coefficients $\alpha_i/p_i(-e_i)$ for any $\alpha_i \in \mathbb{Z}_p$, which yields

$$\sum_{i=1}^q \frac{\alpha_i}{p_i(-e_i)} A_i = \frac{1}{p(x)} \sum_{i=1}^q \left(\alpha_i \frac{p_i(x)}{p_i(-e_i)} G_1 + m_i \alpha_i \frac{p_i(x)}{p_i(-e_i)} H \right).$$

Again, since we can uniquely represent any polynomial as a linear combination of the Lagrange basis, the sum over coefficients of G_1 is $g_0(x)G_1$, where $g_0 \in \mathbb{Z}_p[X]$ is such that $\deg g_0 < q$ and $g_0(-e_i) = \alpha_i$ for all $i \in [q]$. Similarly, the sum over the coefficients of H is $g_1(x)H$, for some polynomial $g_1 \in \mathbb{Z}_p[X]$ such that $\deg g_1 \leq q$ and $g_1(-e_i) = m_i \alpha_i$ for all $i \in [q]$. This induces the constraints that $g_1(-e_i) = m_i g_0(-e_i)$ for $i \in [q]$.

Now, if we can find polynomials g_0 and g_1 of low enough degrees satisfying such constraints, we can construct group elements of the form $x^k \overline{G}_1$ for $0 \leq k < q - \max\{\deg g_0, \deg g_1\}$ with $\overline{G}_1 = p(x)^{-1}(g_0(x)G_1 + g_1(x)H)$. Finally, one can find such g_0 and g_1 of degrees at most $q/2$ via rational function interpolation (see e.g., [SBB⁺80, Theorem 2.2.1.3.])—ultimately, we can find the rational function $r(X) = \frac{g_1(X)}{g_0(X)}$ such that $r(-e_i) = m_i$ for all $i \in [q]$. This then allows us to construct a $(q/2 - 1)$ -DL instance and apply Cheon's attack to extract x .³

Additionally, we consider the case of BBS+ when the messages are different, where the signatures on $m_i \in \mathbb{Z}_p$ are of the form (A_i, e_i, s_i) with

$$A_i = \frac{G_1 + m_i \mathbf{H}[1] + s_i \mathbf{H}[2]}{x + e_i}.$$

Here, the above attack does not work due to the additional term $s_i \mathbf{H}[2]$ with s_i uniformly random (and not chosen by our attack). We show in Section 4 how to extend our ideas to this case and more.

TOWARDS EQUIVALENCE WITH q -SDH. Our above attack can be seen as a reduction from the security of deterministic BBS signatures to the hardness of $(q/2 - 1)$ -DL problem. Although this has direct implications to the security level of the scheme due to Cheon's attack, there is still a qualitative gap between q -DL and q -SDH problem. Moreover, since the security of BBS and BBS+ in the standard model is proved based on q -SDH assumption, we ideally want to establish the equivalence that the security of BBS and BBS+ implies the q -SDH assumption as well. For randomized BBS, such statement is implied by the equivalence of Boneh-Boyen signatures and q -SDH assumption by Jao and Yoshida [JY09]. However, for randomized BBS+ and deterministic BBS, the same equivalence is not known.

We then extend our sketched attack above to this case by first using the $(q/2 - 1)$ -DL instance as a $(q/2 - 1)$ -SDH instance (which is of the same format). Then, given a $(q/2 - 1)$ -SDH adversary $\mathcal{A}$, one can obtain a solution of the form $(e^*, Z^* = \frac{1}{x+e^*} \overline{G}_1)$ with $\overline{G}_1$ as defined before. A key observation is that, applying the properties of Lagrange basis, we can linearly combine Z^* and $(A_i)_{i \in [q]}$ into a group element A^* of the following form:

$$A^* = \frac{g_0(-e^*)G_1 + g_1(-e^*)H}{(x + e^*)p(-e^*)}$$

³ In this sketch, we neglect some edge cases, which we address in Section 4.

Then, we derive $p(-e^*)g_0(-e^*)^{-1} \cdot A^* = \frac{G_1 + m^* H_1}{x + e^*}$ along with e^* as a valid BBS signature for $m^* = g_1(-e^*)g_0(-e^*)^{-1}$. However, this is not the case if $e^* = e_i$ for some $i \in [q]$ or $g_0(-e^*) = 0$, as you cannot find inverse of $p(-e^*)$ and $g_0(-e^*)$ in $\mathbb{Z}_p^*$. Moreover, ruling out these bad events are non-trivial, and this is due to (1) the dependency of the inputs of the adversary $\mathcal{A}$ on the e_i 's and g_0, g_1 , and (2) the structure of polynomials g_0, g_1 which makes the event $g_0(-e^*) = 0$ non-trivial to analyze even when e^* (which is adversarially chosen) is independent from g_0 . We refer to Section 5 for the formal analysis of this sketched reduction.

2 Preliminaries

NOTATIONS. We denote $[n..m] = \{n, n+1, \dots, m\}$ for any integers n, m where $n \leq m$ and $[n] = [1..n]$ for any positive integer n . Throughout the paper, we use λ as the security parameter. Denote $a \leftarrow^* S$ as uniformly sampling a from a finite set S . We denote $y \leftarrow^* \mathcal{A}(x)$ as running a (probabilistic) algorithm $\mathcal{A}$ on input x with fresh randomness. We will denote formal variables in polynomials with sans-serif letters (e.g., X, Y). For any prime modulus p , let $\mathbb{Z}_p[X]$ denote the ring of polynomials containing $g(X) = \sum_{i=0}^d a_i X^i$ with coefficients in $\mathbb{Z}_p$ and with d as the degree of $g(X)$, alternatively denoted using $\deg g$. We might refer to polynomial $g(X)$ using the shorthand g when it is clear from the context. We often denote vectors using bold-sized letters (e.g., $\mathbf{v}, \mathbf{H}$).

LAGRANGE BASIS POLYNOMIALS. We briefly recall the Lagrange basis polynomials. Given distinct $e_1, \dots, e_q \in \mathbb{Z}_p$, let $p_i(X) := \prod_{j \neq i} (X - e_j) \in \mathbb{Z}_p[X]$ for all $i \in [q]$. Then, for any polynomial $g \in \mathbb{Z}_p[X]$ such that $\deg g < q$, $g(X)$ can be uniquely written as a linear combination of the Lagrange basis $\{L_i(X) = p_i(e_i)^{-1} p_i(X)\}_{i \in [q]}$ as $g(X) = \sum_{i=1}^q g(e_i) L_i(X)$.

BILINEAR PAIRING GROUPS. We work with prime order groups. For any such group $\mathbb{G}$ with order p , we denote $0_{\mathbb{G}}$ as the identity element, and $\mathbb{G}^*$ as the set of all $p-1$ generators of $\mathbb{G}$. We adopt additive notations for group elements, and denote group elements with upper-case letters and scalars with lower-case letters. For $G \in \mathbb{G}^*$ and $H \in \mathbb{G}$, we denote $\text{DL}_G(H) \in \mathbb{Z}_p$ as the discrete logarithm of H base G .

A *bilinear group parameter generator* is a probabilistic algorithm GGen taking input 1^λ and outputting $(p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, \mathbf{e})$ such that $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ are groups of λ -bit prime order p , and $\mathbf{e} : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a bilinear map. In particular, the map $\mathbf{e}$ satisfies (1) *bilinearity*, i.e., for any $A \in \mathbb{G}_1, B \in \mathbb{G}_2$ and $x, y \in \mathbb{Z}_p$, $\mathbf{e}(xA, yB) = (xy) \cdot \mathbf{e}(A, B)$, and (2) *non-triviality*, i.e., for any $G_1 \in \mathbb{G}_1^*, G_2 \in \mathbb{G}_2^*$, $\mathbf{e}(G_1, G_2) \in \mathbb{G}_T^*$.

SIGNATURE SCHEMES. A digital signature scheme $\text{SS} = (\text{SS.Setup}, \text{SS.KG}, \text{SS.Sign}, \text{SS.Ver})$ has the following syntax:

- The setup algorithm $\text{SS.Setup}(1^\lambda)$ generates public parameters par which will be used as an implicit input to all other algorithms. We let the public parameters par define the message space $\text{SS.M} = \text{SS.M}(\text{par})$.
- The key generation algorithm $\text{SS.KG}()$ outputs the signing and verification keys (sk, vk) .
- The (possibly randomized) signing algorithm $\text{SS.Sign}(\text{sk}, m)$ takes as inputs, the signing key sk and a message $m \in \text{SS.M}$, and outputs a signature σ .
- The deterministic verification algorithm outputs a bit $\text{SS.Ver}(\text{vk}, m, \sigma)$.

Correctness is defined as usual in that for any public parameters par and key pair (sk, vk) generated from the setup and key generation algorithms and any message $m \in \text{M}$, the signature $\sigma \leftarrow \text{SS.Sign}(\text{sk}, m)$ always satisfies $\text{SS.Ver}(\text{vk}, m, \sigma) = 1$. In this paper, we consider *strong unforgeability* security, which is defined by the game $\text{SUF}_{\text{SS}}^A(\lambda)$ (given in Figure 1). The corresponding advantage of any adversary $\mathcal{A}$ playing the game is given by

$$\text{Adv}_{\text{SS}}^{\text{SUF}}(\mathcal{A}, \lambda) := \Pr[\text{SUF}_{\text{SS}}^A(\lambda) = 1] .$$

SECURITY ASSUMPTIONS. We will use the following variant of *q-Strong Diffie-Hellman* (q -SDH) assumption, as defined by Boneh and Boyen [BB08], in a format supporting type-3 pairings. We will also consider the *q-Discrete Logarithm* (q -DL) assumption, which is implied by the q -SDH assumption. Note that we drop q when it is equal to 1 and refer to it as the *Discrete Logarithm* (DL) assumption. The corresponding games

Game $\text{SUF}_{\text{SS}}^A(\lambda)$: $\text{Sigs} \leftarrow \emptyset$; $\text{par} \leftarrow \text{SS.Setup}(1^\lambda)$ $(\text{sk}, \text{vk}) \leftarrow \text{SS.KG}()$ $(m^*, \sigma^*) \leftarrow \mathcal{A}^{\text{S}}(\text{par}, \text{vk})$ If $(m^*, \sigma^*) \notin \text{Sigs} \wedge \text{SS.Ver}(\text{vk}, m^*, \sigma^*) = 1$ then Return 1 Return 0	Oracle $\text{S}(m)$: $\sigma \leftarrow \text{SS.Sign}(\text{sk}, m)$ If $\sigma \neq \perp$ then $\text{Sigs} \leftarrow \text{Sigs} \cup \{(m, \sigma)\}$ Return σ
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Fig. 1. Strong unforgeability game

Game $\text{q-SDH}_{\text{GGen}}^A(\lambda)$: $\text{par} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GGen}(1^\lambda)$ $G_1 \leftarrow \mathbb{G}_1^*$; $G_2 \leftarrow \mathbb{G}_2^*$; $x \leftarrow \mathbb{Z}_p$ $(e, Z) \leftarrow \mathcal{A}(\text{par}, G_1, (x^i G_1)_{i \in [q]}, G_2, xG_2)$ Return $\left(Z = \frac{1}{x + e} G_1 \right)$	Game $\text{q-DL}_{\text{GGen}}^A(\lambda)$: $\text{par} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GGen}(1^\lambda)$ $G_1 \leftarrow \mathbb{G}_1^*$; $G_2 \leftarrow \mathbb{G}_2^*$; $x \leftarrow \mathbb{Z}_p$ $x' \leftarrow \mathcal{A}(\text{par}, G_1, (x^i G_1)_{i \in [q]}, G_2, xG_2)$ Return $(x = x')$
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Fig. 2. Games q-SDH and q-DL

q-SDH and q-DL are formalized in Figure 2, and we define the corresponding advantages for any adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{GGen}}^{(\text{q-sdh/q-dl})}(\mathcal{A}, \lambda) = \Pr[(\text{q-SDH/q-DL})_{\text{GGen}}^A(\lambda) = 1] .$$

3 BBS Signatures

We give the pseudocode description of the BBS signature scheme in Figure 3 as defined in [TZ23]. The scheme, denoted $\text{BBS} = \text{BBS}[\text{GGen}, \mathcal{D}_e, \ell]$, is parameterized by a bilinear group parameter generator GGen , a distribution $\mathcal{D}_e$ over $\mathbb{Z}_p$, and an integer function $\ell = \ell(\lambda) \geq 1$ denoting the message length (which also defines the message space $\text{BBS.M} = \mathbb{Z}_p^\ell$). We omit $\mathcal{D}_e$ in the parameters when it is clear from the context that e is sampled uniformly from $\mathbb{Z}_p$ (i.e., denoting the scheme as $\text{BBS} = \text{BBS}[\text{GGen}, \ell]$). In the case that $x + e = 0$, we denote $1/0 = 0$ for ease of syntax, following the notations in [TZ23]. Additionally, since this paper considers BBS+ and deterministic BBS signatures, we detail the specific modifications below.

BBS+ SIGNATURES. The $\text{BBS+} = \text{BBS}+[\text{GGen}, \mathcal{D}_e, \ell]$ signature scheme (also given in Figure 3) can be seen as $\text{BBS}[\text{GGen}, \mathcal{D}_e, \ell + 1]$ signature where the first ℓ attributes of the message are determined by the BBS+ message $\mathbf{m} \in \mathbb{Z}_p^\ell$ and the $(\ell + 1)$ -th attribute, denoted s , is uniformly sampled from $\mathbb{Z}_p$ during signing. In particular, the setup and key generation are exactly that of BBS's. The signing algorithm outputs a signature of the form (A, e, s) , and the verification of $(\mathbf{m} \in \mathbb{Z}_p^\ell, (A, e, s))$ can be seen as BBS verification of $((\mathbf{m}, s) \in \mathbb{Z}_p^{\ell+1}, (A, e))$.

DETERMINISTIC BBS. For the deterministic variant of BBS, instead of sampling e from $\mathcal{D}_e$ during signing, e is computed by evaluating a pseudorandom function $F(\text{seed}, \cdot)$ on the message $\mathbf{m}$. This seed is picked during key generation and appended to the secret key. One can also define a deterministic variant of BBS+ accordingly, where both (e, s) are generated via a PRF.

PAIRING-FREE VARIANT. The BBS and BBS+ signature scheme can also be casted as an algebraic message authentication codes (MAC) as done in [BBDT16, Orr24]. In particular, the scheme relies instead on pairing-free groups and the verification is done using the secret key to check the format of the MAC tag (which is simply a BBS/BBS+ signature) instead of the public key. We note that our attacks do not rely on the existence of bilinear pairings and also apply to these schemes.

Algorithm BBS.Setup(1^λ) : $(p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GGen}(1^\lambda)$ $G_1 \leftarrow \mathbb{G}_1^*$; $G_2 \leftarrow \mathbb{G}_2^*$ $\boxed{H \leftarrow \mathbb{G}_1^\ell}$ $\boxed{H \leftarrow \mathbb{G}_1^{\ell+1}}$ $\text{par} \leftarrow (p, G_1, \mathbf{H}, G_2, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e)$ Return par Algorithm BBS.Ver($\text{vk}, \mathbf{m}, \sigma$) : $(A, e) \leftarrow \sigma$ $C \leftarrow G_1 + \sum_{i=1}^\ell \mathbf{m}[i] \mathbf{H}[i]$ $\boxed{(A, e, s) \leftarrow \sigma}$ $\boxed{C \leftarrow G_1 + s \mathbf{H}[\ell+1] + \sum_{i=1}^\ell \mathbf{m}[i] \mathbf{H}[i]}$ Return $\mathbf{e}(A, \text{vk} + e G_2) = \mathbf{e}(C, G_2)$	Algorithm BBS.KG() : $(p, G_1, \mathbf{H}, G_2, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{par}$ $x \leftarrow \mathbb{Z}_p$ Return $(\text{sk} \leftarrow x, \text{vk} \leftarrow x G_2)$ Algorithm BBS.Sign($\text{sk} = x, \mathbf{m} \in \mathbb{Z}_p^\ell$) : $e \leftarrow \mathcal{D}_e$; $A \leftarrow \frac{1}{x+e} \left(G_1 + \sum_{i=1}^\ell \mathbf{m}[i] \mathbf{H}[i] \right)$ Return (A, e) Algorithm BBS+.Sign($\text{sk} = x, \mathbf{m} \in \mathbb{Z}_p^\ell$) : $e \leftarrow \mathcal{D}_e$; $s \leftarrow \mathbb{Z}_p$ $A \leftarrow \frac{1}{x+e} \left(G_1 + s \mathbf{H}[\ell+1] + \sum_{i=1}^\ell \mathbf{m}[i] \mathbf{H}[i] \right)$ Return (A, e, s)
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Fig. 3. BBS Signature. The signature schemes $\text{BBS} = \text{BBS}[\text{GGen}, \mathcal{D}_e, \ell]$ and $\text{BBS+} = \text{BBS+}[\text{GGen}, \mathcal{D}_e, \ell]$ are parameterized by GGen , a distribution $\mathcal{D}_e$ from which e is sampled, and the message length $\ell = \ell(\lambda) \geq 1$. We denote the description exclusive to BBS and BBS+ in solid and dashed boxes, respectively.

4 Attacks on Deterministic BBS and BBS+ Signatures

In this section, we give our attacks on the strong unforgeability of deterministic BBS and randomized BBS+ signatures using $O(\sqrt{p/q})$ group exponentiations, along with some of their extensions. We state useful lemmas and the main theorem in Section 4.1 with the proofs given in Sections 4.2 to 4.4.

4.1 Useful Lemmas and Main Theorem

Our attacks will crucially rely on the following technical lemmas. Their proofs are given in Sections 4.3 and 4.4. Note that the integer L in the following lemmas is *not necessarily* the message length ℓ . For the case $L = 1$, Lemma 4.1 is already implied by rational function interpolation (see [SBB⁺80, Theorem 2.2.1.3.]). However, we require the lemma for $L > 1$ for our attack on deterministic BBS+ and other extensions.

Lemma 4.1. *Let p be a prime, and L and $\mathbf{q}$ be positive integers such that $(L+1) \mid \mathbf{q}$. Also, let T_p denote the runtime for one field operation in $\mathbb{Z}_p$ and $\mathbf{M}(\mathbf{q})$ be the number of $\mathbb{Z}_p$ -operations for degree $\mathbf{q}$ $\mathbb{Z}_p$ -polynomial multiplication (i.e., $O(\mathbf{q}^2)$ naively and $O(\mathbf{q} \log \mathbf{q})$ if we assume fast multiplication). Then, there exists an algorithm FindPolys which takes as input $p, L, \mathbf{q}$, vectors $\mathbf{m}_1, \dots, \mathbf{m}_{\mathbf{q}} \in \mathbb{Z}_p^L$ and distinct $e_1, \dots, e_{\mathbf{q}} \in \mathbb{Z}_p$, such that it runs in time $O(L^3 \mathbf{M}(\mathbf{q}) \log \mathbf{q} T_p)$ and outputs polynomials $g_0, \dots, g_L \in \mathbb{Z}_p[X]$ where*

- (1) $g_0(X) \neq 0$
- (2) $\deg g_j \leq \mathbf{q}L/(L+1)$ for all $j \in [0..L]$, and
- (3) $g_j(-e_i) = \mathbf{m}_i[j] \cdot g_0(-e_i)$ for all $i \in [\mathbf{q}]$ and $j \in [L]$.

Additionally, one can also uniformly sample $g_0, \dots, g_L$ conditioning on (2) and (3) within running time $O(L^3 \mathbf{q}^3 T_p)$, we denote this algorithm as SampPolys.

Lemma 4.2 (Linear Combination of Rational Functions.). *Let $e_1, \dots, e_{\mathbf{q}} \in \mathbb{Z}_p$ be all distinct. For any $\mathbf{m}_1, \dots, \mathbf{m}_{\mathbf{q}} \in \mathbb{Z}_p^L$ and polynomials $g_0, \dots, g_L \in \mathbb{Z}_p[X]$ of degrees at most $\mathbf{q} - 1$, such that $g_j(-e_i) = \mathbf{m}_i[j] \cdot g_0(-e_i)$ for all $i \in [\mathbf{q}]$ and $j \in [L]$, we have that, for any $0 \leq k < \mathbf{q} - \max_{j \in [0..L]} \{\deg g_j\}$,*

$$\sum_{i=1}^{\mathbf{q}} \frac{(-e_i)^k g_0(-e_i)}{p_i(-e_i)} \left(\frac{1 + \sum_{j=1}^L \mathbf{m}_i[j] Y_j}{X + e_i} \right) = \frac{X^k (g_0(X) + \sum_{j=1}^L g_j(X) Y_j)}{p(X)},$$

where $p(X) = \prod_{i=1}^q (X + e_i)$ and $p_i(X) = \prod_{j \neq i} (X + e_j)$ for all $i \in [q]$.

Now, we state our main theorem, proved in Section 4.2, which shows that breaking strong unforgeability of BBS and BBS+ signatures, after making q signing queries on *different messages*, can be reduced to solving a $\Theta(q)$ -DL instance. We first focus on BBS and BBS+ signatures with message length $\ell = 1$.

Theorem 4.3. *Let GGen be a bilinear group parameter generator producing groups of prime order $p = p(\lambda)$, $\mathcal{D}_e$ be a distribution suitable for BBS and BBS+ signatures, and $q' = q'(\lambda)$ be a positive integer. Also, let $T = T(\lambda)$ and $T_p = T_p(\lambda)$ denote the maximum time used for one exponentiation in $\mathbb{G}_1$ and one field operation in $\mathbb{Z}_p$, respectively, and $M(q)$ be the number of $\mathbb{Z}_p$ -operations for degree q $\mathbb{Z}_p$ -polynomial multiplication. For any q' -DL adversary $\mathcal{A}$ running in time $t_{\mathcal{A}} = t_{\mathcal{A}}(\lambda)$, there exist*

1. *An adversary $\mathcal{A}_1$ playing the SUF game of $\text{BBS} = \text{BBS}[\text{GGen}, \mathcal{D}_e, 1]$ such that*

$$\text{Adv}_{\text{BBS}}^{\text{suf}}(\mathcal{A}_1, \lambda) \geq \text{Adv}_{\text{GGen}}^{q'\text{-dl}}(\mathcal{A}, \lambda) - \frac{1}{p}.$$

Furthermore, $\mathcal{A}_1$ makes $q = 2q' + 2$ signing queries with different messages and runs in time roughly $t_{\mathcal{A}} + O(q^2 T + M(q) \log q T_p)$.

2. *An adversary $\mathcal{A}_2$ playing the SUF game of $\text{BBS+} = \text{BBS+}[\text{GGen}, \mathcal{D}_e, 1]$ such that*

$$\text{Adv}_{\text{BBS+}}^{\text{suf}}(\mathcal{A}_2, \lambda) \geq \text{Adv}_{\text{GGen}}^{q'\text{-dl}}(\mathcal{A}, \lambda) - \frac{1}{p}.$$

Furthermore, $\mathcal{A}_2$ makes $q = 3q' + 3$ signing queries with different messages and runs in time roughly $t_{\mathcal{A}} + O(q^2 T + M(q) \log q T_p)$.

We recall Cheon's attack [Che06] in the following theorem which, combined with the above reductions, yields attacks against BBS and BBS+ using $O(\sqrt{p/q})$ group exponentiations, stated in Corollary 4.5.

Theorem 4.4 (Cheon's Attack [Che06]). *Let $\mathbb{G}$ be a cyclic group of prime order $p = p(\lambda)$ with a generator G . Also, let $T = T(\lambda)$ denote the maximum time used for one exponentiation in $\mathbb{G}$.*

1. *If $d \mid (p-1)$, given group elements of the form $G, xG, x^d G$, there exists a deterministic algorithm that finds $x \in \mathbb{Z}_p$ in time $O((\sqrt{p/d} + \sqrt{d})T)$.*
2. *If $d \mid (p+1)$, given group elements of the form $G, xG, \dots, x^{2d}G$, there exists a deterministic algorithm that finds $x \in \mathbb{Z}_p$ in time $O((\sqrt{p/d} + d)T)$.*

Corollary 4.5 (Attacks on BBS and BBS+). *Let GGen , p , T , and T_p be as defined in Theorem 4.3. Let d be a positive integer such that $d \mid (p-1)$ or $d \mid (p+1)$. Then, there exist adversaries $\mathcal{A}_1$ and $\mathcal{A}_2$ playing the SUF security game of BBS and BBS+, respectively, both with success probability $1 - 1/p$. Both adversaries make $q = \Theta(d)$ signing queries on different messages and run in time roughly $O((\sqrt{p/d} + d^2)T + M(d) \log d T_p)$.*

The above attack $\mathcal{A}_1$ immediately implies an attack on deterministic BBS signatures, since the signing queries are made on distinct messages. Moreover, if one were to consider a deterministic variant of BBS+, the attack $\mathcal{A}_2$ would apply with the same reasoning. We also make some remarks on possible extensions.

Remark 4.6. One could reduce the running time further to $O((\sqrt{p/d} + d)T + M(d) \log d T_p)$ when $d \mid (p-1)$ as Cheon's attack only requires computing $G, xG, x^d G$ instead of $G, \dots, x^d G$. For specific bilinear curves allowing FFT in the field $\mathbb{Z}_p$, e.g., BLS12-381 with several divisors $d \mid p-1$, $q = O(q \log q)$, making the overall runtime of our attacks $O((\sqrt{p/d} + d)T + d \log^2 d T_p)$ with $\sqrt{p/d}T$ as the dominating term.

Remark 4.7. Our attacks can easily be extended to BBS and BBS+ (and their deterministic variants) with message space $\mathbb{Z}_p^\ell$ without any complexity loss. By setting each message to be $(m_i, 0, \dots, 0)$ for distinct $m_i \in \mathbb{Z}_p$, each BBS signature (A_i, e_i) will be of the form $A_i = (x + e_i)^{-1}(G_1 + m_i \mathbf{H}[1])$, allowing attack $\mathcal{A}_1$ to still apply. A similar extension applies to attack $\mathcal{A}_2$ for BBS+.

Remark 4.8. Even more generally, we can also extend the attacks on BBS (and BBS+) for any messages $\mathbf{m} \in \mathbb{Z}_p^\ell$. In this case, we can generalize Theorem 4.3 to give a reduction from SUF security to the q' -DL assumption that makes $q = (\ell + 1)(q' + 1)$ (and $(\ell + 2)(q' + 1)$ for BBS+) signing queries on any different messages in $\mathbb{Z}_p^\ell$. Note the incurred loss in the signing query count.

Adversary $\mathcal{A}_1^S(\text{par}, \text{vk})$ $\mathcal{A}_2^S(\text{par}, \text{vk})$: $((p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e), G_1, H, G_2) \leftarrow \text{par}$ $((p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e), G_1, (H[1], H[2]), G_2) \leftarrow \text{par}$ If $\text{vk} = 0_{\mathbb{G}_2}$ then $x \leftarrow 0$ $q \leftarrow 2q' + 2$ $q \leftarrow 3q' + 3$ Select any distinct $m_1, \dots, m_q \in \mathbb{Z}_p$ For $i \in [q]$: $(A_i, e_i, s_i) \leftarrow \mathbf{S}(m_i)$ If $\exists j < i : e_j = e_i$ then Return $\left(\frac{m_i + m_j}{2}, \left(\frac{A_i + A_j}{2}, e_i, \left\lfloor \frac{s_i + s_j}{2} \right\rfloor \right) \right)$ $(g_0, g_1) \leftarrow \text{FindPolys}(p, 1, q, (m_i, e_i)_{i \in [q]})$ $(g_0, g_1, g_2) \leftarrow \text{FindPolys}(p, 2, q, ((m_i, s_i), e_i)_{i \in [q]})$ $r \leftarrow \mathbb{Z}_p^*$ For $i \in [q]$: $\alpha_i \leftarrow g_0(-e_i)/p_i(-e_i)$ For $k \in [0..q']$: $X_{1,k} \leftarrow r \sum_{i=1}^q (-e_i)^k \alpha_i A_i$ If $x = \perp \wedge X_{1,0} = 0_{\mathbb{G}_1}$ then If $\exists \hat{x} \in \mathbb{Z}_p : \hat{x}G_2 = \text{vk} \wedge 0 = g_0(\hat{x}) = g_1(\hat{x})$ then $x \leftarrow \hat{x}$ Else abort If $\exists i \in [q] : \text{vk} = -e_i G_2$ then $x \leftarrow -e_i$ If $x = \perp$ then $x \leftarrow \mathcal{A}(\text{par}, X_{1,0}, (X_{1,k})_{k \in [q']}, G_2, \text{vk})$ Select $m^* \notin \{m_1, \dots, m_q\}$ and $e^* \neq -x$ Return $(m^*, ((x + e^*)^{-1}(G_1 + m^*H), e^*))$ Return $(m^*, ((x + e^*)^{-1}(G_1 + m^*H[1]), e^*, 0))$	Algorithm $\text{FindPolys}(p, L, q, (\mathbf{m}_i, e_i)_{i \in [q]})$: $n \leftarrow qL/(L+1)$ For $j \in [L]$: Find $h_j(X)$ s.t. $h_j(-e_i) = \mathbf{m}_i[j], \forall i \in [q]$ $\alpha(X) \leftarrow \prod_{i=1}^q (X + e_i)$ $(g_0, \dots, g_L) \leftarrow \text{VectorRecon}(\mathbf{h}, \alpha, n, n)$ Return $(g_0, \dots, g_L)$ Algorithm $\text{SampPolys}(p, L, q, (\mathbf{m}_i, e_i)_{i \in [q]})$: $n \leftarrow qL/(L+1)$ For $j \in [L]$: $\mathbf{M}^{(j)} \leftarrow \text{Diag}(\mathbf{m}_1[j], \dots, \mathbf{m}_q[j])$ $\mathbf{V} \leftarrow ((-e_i)^k)_{(i,k) \in [q] \times [0..n]}$ $\mathbf{A} \leftarrow \begin{pmatrix} -\mathbf{M}^{(1)} \mathbf{V} & \mathbf{V} \\ \vdots & \ddots \\ -\mathbf{M}^{(L)} \mathbf{V} & \mathbf{V} \end{pmatrix}$ $\mathbf{A} \in \mathbb{Z}_p^{qL \times (n+1)(L+1)}$ Sample $\mathbf{b} \leftarrow \ker(\mathbf{A})$ For $i \in [0..L]$: $g_i(X) \leftarrow \sum_{j=0}^n b_j^{(i)} X^j$ Return $(g_0, \dots, g_L)$
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Fig. 4. Description of adversaries $\mathcal{A}_1$ and $\mathcal{A}_2$ in Theorem 4.3. The description of $\mathcal{A}_1$ includes everything but the dashed boxes, while the description of $\mathcal{A}_2$ includes everything but the solid boxes. Both adversaries run a q' -DL adversary $\mathcal{A}$ as a subroutine. The description of FindPolys is given in the right-hand side, along with an alternative randomized version SampPolys . The algorithm VectorRecon is the vector rational function reconstruction solver in [OS06] as discussed in Section 4.3. We denote the pseudocode of SampPolys in the dashed box. The shorthand $\text{Diag}(a_1, \dots, a_q)$ denotes a $q \times q$ diagonal matrix with diagonal entries $a_1, \dots, a_q$.

4.2 Proof of Theorem 4.3

The adversaries $\mathcal{A}_1$ and $\mathcal{A}_2$ playing the SUF game of BBS and BBS+ signatures, respectively, are described in Figure 4. They follow the following template:

- (1) Select any distinct $\mathbf{m}_1, \dots, \mathbf{m}_q$ (e.g., $\mathbf{m}_i = i$ for $i \in [q]$) and query for signatures of these messages.
- (2) Use the algorithm FindPolys from Lemma 4.1 to find polynomials $g_0, \dots, g_L$ ($L = 1$ for BBS and $L = 2$ for BBS+) of degrees at most $qL/(L+1)$ such that $g_j(-e_i) = \mathbf{m}_i[j]g_0(-e_i)$ for all $i \in [q]$ and $j \in [L]$. Note here that for $\mathcal{A}_2$, we redefine $\mathbf{m}_i = (\mathbf{m}_i, s_i)$.
- (3) By Lemma 4.2, we can linearly combine the signatures to compute a q' -DL instance $X_{1,k} = x^k X_{1,0}$ for $k \in [0..q']$ with x denoting the secret key. Then, we run $\mathcal{A}$ on this instance to find x , which we use to forge a signature.

The running time of $\mathcal{A}_i$ for $i \in \{1, 2\}$ includes the running time of $\mathcal{A}$ and FindPolys , and the cost of computing $X_{1,k}$ for $k \in [0..q']$.⁴ Since each $X_{1,k}$ costs q exponentiation to compute, $\mathcal{A}_i$ runs in time roughly $t_A + O(q^2 T + M(q) \log q T_p)$. We then analyze the success probabilities of $\mathcal{A}_1$ and $\mathcal{A}_2$.

⁴ Note that if $\mathcal{A}$ only needs $X_{1,0}, X_{1,1}, X_{1,q'}$, then Step (3) only takes $O(q)$ exponentiations instead of $O(q^2)$.

ANALYSIS OF $\mathcal{A}_1$. We first rule out the case that $x = 0$ or $x \in \{-e_1, \dots, -e_q\}$, in which $\mathcal{A}_1$ can easily detect and output a forgery. Then, to apply Lemmas 4.1 and 4.2 for $L = 1$, we want to ensure that $e_1, \dots, e_q$ are all distinct.

Indeed, if $e_i = e_j$ for some $i \neq j \in [q]$, $\mathcal{A}_1$ can successfully forges a new signature $(A^* = (A_i + A_j)/2, e_i)$ of the message $m^* = (m_i + m_j)/2$. It is easy to see that this is a valid signature.

Then, by Lemma 4.1, FindPolys outputs polynomials $g_0, g_1 \in \mathbb{Z}_p[X]$ of degrees at most $q/2 = q' + 1$ such that g_0 is non-zero, and $g_1(-e_i) = m_i g_0(-e_i)$ for all $i \in [q]$. With $\alpha_i = g_0(-e_i)/p(-e_i)$ for $i \in [q]$, we consider the following linear combination of A_i 's for $k \in [0..q']$,

$$\begin{aligned} X_{1,k} &= \sum_{i=1}^q (-e_i)^k \alpha_i A_i = \sum_{i=1}^q \frac{(-e_i)^k g_0(-e_i)}{p(-e_i)} \left(\frac{G_1 + m_i H}{x + e_i} \right) \\ &= \frac{x^k (g_0(x) G_1 + g_1(x) H)}{p(x)}. \end{aligned}$$

The last equality follows from Lemma 4.2 setting $X = x$ and $Y_1 = \text{DL}_{G_1}(H)$ and that $\deg X^{q'} g_0(X) \leq 2q' + 1 < q$. In the case that $X_{1,0} = 0_{G_1}$, we know that either the secret key x is a common root of g_0 and g_1 or $g_0(x) G_1 = -g_1(x) H$ with $g_1(x) \neq 0$. In the former case, we can efficiently find x by factoring g_0, g_1 (since g_0 is non-zero). The latter case, which we will denote as event **BadG**, occurs with $1/p$ probability, since H is uniform in G_1 and independent of x and e_i 's. Otherwise, $X_{1,0}$ is a uniformly random generator (as it is rerandomized with $r \leftarrow \mathbb{Z}_p^*$), and $X_{1,k} = x^k X_{1,0}$ for $k \in [q']$.

Since $\mathcal{A}_1$ only aborts when **BadG** occur and wins if $\mathcal{A}$ successfully computes x , $\mathcal{A}_1$ wins with probability at least $\text{Adv}_{\text{GGen}}^{q'-\text{dl}}(\mathcal{A}, \lambda) - 1/p$.

ANALYSIS OF $\mathcal{A}_2$. The analysis for $\mathcal{A}_2$ follows similarly from that of $\mathcal{A}_1$ except that we apply Lemmas 4.1 and 4.2 with $L = 2$, and the inputs to FindPolys are now $\mathbf{m}_i = (m_i, s_i)$ for all $i \in [q]$. Note that if for some $i \neq j, e_i = e_j$, $\mathcal{A}_2$ can output a valid signature $(A^* = (A_i + A_j)/2, e_i, s^* = (s_i + s_j)/2)$ for the message $m^* = (m_i + m_j)/2$. Hence, we only consider when $e_1, \dots, e_q$ are distinct.

Then, by Lemma 4.1 with $L = 2$, FindPolys outputs polynomials $g_0 \neq 0, g_1, g_2 \in \mathbb{Z}_p[X]$ of degrees at most $2q/3 = 2q' + 2$ such that $g_1(-e_i) = m_i g_0(-e_i)$ and $g_2(-e_i) = s_i g_0(-e_i)$ for all $i \in [q]$.

Next, we consider the group elements $X_{1,k} = \sum_{i=1}^q (-e_i)^k g_0(-e_i)/p_i(-e_i) A_i$ for $k \in [0..q']$. By Lemma 4.2, setting $X = x$ and $Y_i = \text{DL}_{G_1} \mathbf{H}[i]$ for $i = 1, 2$, and that $\deg X^{q'} g_0(X) \leq 3q' + 2 < q$, we have that for $k \in [0..q']$,

$$X_{1,k} = \frac{x^k (g_0(x) G_1 + g_1(x) \mathbf{H}[1] + g_2(x) \mathbf{H}[2])}{p(x)}.$$

Again, if $X_{1,0} = 0_{G_1}$, then x is a common root of g_0, g_1 and g_2 , which can be computed by factoring the polynomials, or a non-zero linear combination of $G_1, \mathbf{H}[1], \mathbf{H}[2]$ is 0_{G_1} (similarly denoted as event **BadG** which occurs with $1/p$ probability, since these generators are uniformly random and independent of x and e_i 's). Otherwise, $X_{1,0}$ is a uniformly random generator (as it is rerandomized with $r \leftarrow \mathbb{Z}_p^*$), and $\mathcal{A}_2$ wins if $\mathcal{A}$ successfully computes x . Thus, $\mathcal{A}_2$ wins with probability at least $\text{Adv}_{\text{GGen}}^{q'-\text{dl}}(\mathcal{A}, \lambda) - 1/p$. $\square$

4.3 Proof of Lemma 4.1

Let $n = qL/(L+1)$. We first prove the existence of the polynomials satisfying the lemma statement and refer to Figure 4 for the description of FindPolys and SampPolys. Consider $L+1$ polynomials $g_0, \dots, g_L \in \mathbb{Z}_p[X]$, each with degree at most n , denoted as $g_j(X) := \sum_{k=0}^n b_k^{(j)} X^k$ for $j \in [0..L]$. Then, the existence of $g_0, \dots, g_L$, such that $g_j(-e_i) = \mathbf{m}_i[j] g_0(-e_i)$ for all $i \in [q]$ and $j \in [L]$, depends on whether the following system of linear equations on the variables $(b_0^{(0)}, \dots, b_n^{(0)}, \dots, b_0^{(L)}, \dots, b_n^{(L)})$ ⁵

$$\sum_{k=0}^n (-e_i)^k b_k^{(j)} - \mathbf{m}_i[j] \sum_{k=0}^n (-e_i)^k b_k^{(0)} = 0, \text{ for all } j \in [L] \text{ and } i \in [q],$$

⁵ Denoted with sans-serif fonts indicating that they are formal variables.

has a solution. This linear equation corresponds to the equation $\mathbf{A}\mathbf{b} = \mathbf{0}$ where the matrix $\mathbf{A} \in \mathbb{Z}_p^{\mathbf{q}L \times (n+1)(L+1)}$ is as described in **FindPolys**.

Since $\mathbf{A}$ is a linear map from $\mathbb{Z}_p^{(n+1)(L+1)}$ to $\mathbb{Z}_p^{L\mathbf{q}}$ and $(n+1)(L+1) = L\mathbf{q} + L + 1 > L\mathbf{q}$, the kernel of $\mathbf{A}$ has dimension at least 1. Thus, there exists a non-trivial solution to this system of equations, denoted $\hat{\mathbf{b}} = (\hat{b}_0^{(0)}, \dots, \hat{b}_n^{(0)}, \dots, \hat{b}_n^{(L)})^T \neq \mathbf{0}$. Then, it is left to show that $\hat{\mathbf{b}}$ induces a non-zero polynomial g_0 .

Suppose $g_0(\mathbf{X}) = 0$. Then, for all $j \in [L]$ and $i \in [\mathbf{q}]$, $g_j(-e_i) = 0$. With the e_i 's distinct, g_j has $\mathbf{q}$ roots. Since $\deg g_j \leq n < \mathbf{q}$, $g_j(\mathbf{X}) = 0$ for all $j \in [L]$, which is a contradiction with $\hat{\mathbf{b}} \neq \mathbf{0}$. We note that although such non-trivial solution can be found in $O(\mathbf{q}^3 L^3)$ operations in $\mathbb{Z}_p$ via Gaussian elimination, there exists a more efficient algorithm to find such polynomials which we discuss below. Note that to efficiently sample the coefficients $\mathbf{b}$ of $g_0, \dots, g_L$ which is in the kernel of $\mathbf{A}$, we can apply Gaussian elimination on $\mathbf{A}$, reducing it to a row-echelon form (i.e., a matrix where the first non-zero columns for each row are distinct), and sample the coefficients that does not correspond to a leading coefficients of each row.

Instead, we can view this polynomial finding problem as a vector rational function reconstruction problem [OS06]. Here, we let $\alpha(\mathbf{X}) := \prod_{i=1}^{\mathbf{q}} (\mathbf{X} + e_i)$ and $h_j(\mathbf{X})$ for each $j \in [L]$ be the degree at most $\mathbf{q} - 1$ polynomial such that $h_j(-e_i) = \mathbf{m}_i[j]$ for all $i \in [\mathbf{q}]$. (Note that h_j can be found by polynomial interpolation, which takes $O(\mathbf{M}(\mathbf{q}) \log \mathbf{q} T_p)$ time.) The problem is then defined as follows: Given integers N, D and polynomials $\alpha, h_1, \dots, h_L$ with $\deg \alpha > N, D$, find $g_0, g_1, \dots, g_L$ such that for each $j \in [L]$

$$g_0(\mathbf{X})h_j(\mathbf{X}) = g_j(\mathbf{X}) \pmod{\alpha(\mathbf{X})}, \text{ and } \deg g_0 \leq D, \deg g_j \leq N.$$

It is easy to check that such $g_0, \dots, g_L$ satisfy our desired system of linear equations, as $\alpha(-e_i) = 0$ for all $i \in [\mathbf{q}]$ and $h_j(-e_i) = \mathbf{m}_i[j]$ for all i, j .

From [OS06, Corollary 4.1 and Theorem 4.3], there exists an algorithm **VectorRecon**($\mathbf{h}(\mathbf{X}), \alpha, N, D$) which outputs a vector of polynomials of length L containing a polynomial β where $\deg \beta \leq D$ and $\beta(\mathbf{X})h_j(\mathbf{X}) \pmod{\alpha(\mathbf{X})}$ has degree at most N for all $j \in [L]$. In particular, setting $g_0 = \beta$ and $g_j = \beta g_j \pmod{\alpha}$ yields our desired solution. Note that we will call **VectorRecon** as the algorithm which finds our solution instead. Here, the algorithm runs in time $O(L \cdot k^2 \mathbf{M}(\mathbf{q}) \log \mathbf{q})$ where k is a minimal integer such that $\mathbf{q} = \deg \alpha > N + D/k$. With $N = D = \mathbf{q}L/(L+1)$ for us, we have $k = L+1 = O(L)$. Combining with the runtime for L polynomial interpolations for h_j and computing $g_j = \beta h_j \pmod{\alpha}$ (both of which takes $O(L\mathbf{M}(\mathbf{q}) \log \mathbf{q} T_p)$) results in the claimed runtime in the lemma statement. Note that in the worst-case with naive polynomial multiplication this is $O(L^3 \mathbf{q}^2 \log \mathbf{q} T_p)$, which is still better than $O(L^3 \mathbf{q}^3 T_p)$ Gaussian elimination. $\square$

4.4 Proof of Lemma 4.2

Let $e_1, \dots, e_{\mathbf{q}} \in \mathbb{Z}_p$ be all distinct and $p(\mathbf{X})$ and $p_i(\mathbf{X})$ be as in the lemma statement. Consider $\mathbf{m}_1, \dots, \mathbf{m}_{\mathbf{q}} \in \mathbb{Z}_p^L$ and polynomials $g_0, \dots, g_L \in \mathbb{Z}_p[\mathbf{X}]$, such that $\deg g_j < \mathbf{q}$ for all $j \in [0..L]$ and $g_j(-e_i) = \mathbf{m}_i[j] \cdot g_0(-e_i)$ for all $i \in [\mathbf{q}]$ and $j \in [L]$. Let k be an integer in the range $0 \leq k < \mathbf{q} - \max_{j \in [0..L]} \{\deg g_j\}$. Since $\max_{j \in [0..L]} \{\deg \mathbf{X}^k g_j(\mathbf{X})\} \leq k + \max_j \{\deg g_j\} < \mathbf{q}$, we can write $\mathbf{X}^k g_j(\mathbf{X})$ for $j \in [0..L]$ as the following linear combination of the Lagrange basis polynomials

$$\begin{aligned} \mathbf{X}^k g_0(\mathbf{X}) &= \sum_{i=1}^{\mathbf{q}} \frac{(-e_i)^k g_0(-e_i)}{p_i(-e_i)} p_i(\mathbf{X}), \text{ and} \\ \mathbf{X}^k g_j(\mathbf{X}) &= \sum_{i=1}^{\mathbf{q}} \frac{(-e_i)^k g_j(-e_i)}{p_i(-e_i)} p_i(\mathbf{X}) \\ &= \sum_{i=1}^{\mathbf{q}} \mathbf{m}_i[j] \cdot \frac{(-e_i)^k g_0(-e_i)}{p_i(-e_i)} p_i(\mathbf{X}), \text{ for all } j \in [L]. \end{aligned}$$

Therefore, with $\alpha_i = (-e_i)^k g_0(-e_i)/p_i(-e_i)$ for all $i \in [\mathbf{q}]$, we have

$$\begin{aligned} \sum_{i=1}^{\mathbf{q}} \alpha_i \left(\frac{1 + \sum_{j=1}^L \mathbf{m}_i[j] \mathbf{Y}_j}{\mathbf{X} + e_i} \right) &= \frac{1}{p(\mathbf{X})} \left(\sum_{i=1}^{\mathbf{q}} \alpha_i p_i(\mathbf{X}) + \sum_{j=1}^L \sum_{i=1}^{\mathbf{q}} \mathbf{m}_i[j] \alpha_i p_i(\mathbf{X}) \mathbf{Y}_j \right) \\ &= \frac{\mathbf{X}^k (g_0(\mathbf{X}) + \sum_{j=1}^L g_j(\mathbf{X}) \mathbf{Y}_j)}{p(\mathbf{X})}. \quad \square \end{aligned}$$

5 Extending to $\mathbf{q}$ -SDH

In this section, we extend the attack in Section 4 to a reduction from security of BBS and BBS+ to the hardness of $\Theta(\mathbf{q})$ -SDH problem. We first state our main theorem in Section 5.1 along with discussion on extensions of the reduction, and give the proof in Section 5.2

5.1 Reduction from SUF Security to $\Theta(\mathbf{q})$ -SDH

In this section, we show that the reductions $\mathcal{A}_1$ and $\mathcal{A}_2$ from Theorem 4.3 can be extended to reductions $\mathcal{B}_1$ and $\mathcal{B}_2$, breaking SUF security of BBS and BBS+ (making signing $\mathbf{q}$ queries on *any distinct messages*) can be reduced to solving a $\Theta(\mathbf{q})$ -SDH instance. However, this comes with much more complex analysis, which we give in Section 5.2.

Theorem 5.1. *Let GGen be a bilinear group parameter generator producing groups of prime order $p = p(\lambda)$, and $\mathbf{q}' = \mathbf{q}'(\lambda)$ be a positive integer. Also, let $T = T(\lambda)$ and $T_p = T_p(\lambda)$ denote the maximum time used for one exponentiation in $\mathbb{G}_1$ and one field operation in $\mathbb{Z}_p$, respectively. For any $\mathbf{q}'$ -SDH adversary $\mathcal{A}$ running in time $t_{\mathcal{A}} = t_{\mathcal{A}}(\lambda)$, there exist*

1. *An adversary $\mathcal{B}_1$ playing the SUF security game of $\text{BBS} = \text{BBS}[\text{GGen}, 1]$ such that*

$$\text{Adv}_{\text{BBS}}^{\text{suf}}(\mathcal{B}_1, \lambda) \geq \text{Adv}_{\text{GGen}}^{\mathbf{q}'\text{-sdh}}(\mathcal{A}, \lambda) - \frac{5\mathbf{q}^2 + 4\mathbf{q} + 10}{2p}.$$

Furthermore, $\mathcal{B}_1$ makes $\mathbf{q} = 2\mathbf{q}' + 2$ signing queries with different messages and runs in time roughly $t_{\mathcal{A}} + O(\mathbf{q}^2 T + \mathbf{q}^3 T_p)$.

2. *An adversary $\mathcal{B}_2$ playing the SUF security game of $\text{BBS+} = \text{BBS+}[\text{GGen}, 1]$ such that*

$$\text{Adv}_{\text{BBS+}}^{\text{suf}}(\mathcal{B}_2, \lambda) \geq \text{Adv}_{\text{GGen}}^{\mathbf{q}'\text{-sdh}}(\mathcal{A}, \lambda) - \frac{4\mathbf{q}^2 + 11\mathbf{q} + 5}{p}.$$

Furthermore, $\mathcal{B}_2$ makes $\mathbf{q} = 3\mathbf{q}' + 3$ signing queries with different messages and runs in time roughly $t_{\mathcal{A}} + O(\mathbf{q}^2 T + \mathbf{q}^3 T_p)$.

Consequently, we have that SUF security of deterministic BBS and BBS+ implies the $\Theta(\mathbf{q})$ -SDH assumption. We emphasize that the reduction $\mathcal{B}_1$ and $\mathcal{B}_2$ works for any sequence of distinct messages $m_1, \dots, m_{\mathbf{q}}$ to be queried to the signing oracles. This also allows the extension mentioned in Remark 4.7 to apply here.

5.2 Proof of Theorem 5.1

The reductions $\mathcal{B}_1$ and $\mathcal{B}_2$ playing the SUF game of BBS and BBS+ signatures, respectively, are described in Figure 5. They follow the following template:

- (1) Select any distinct $m_1, \dots, m_{\mathbf{q}}$ (e.g., $m_i = i$ for $i \in [\mathbf{q}]$) and query for signatures of these messages.
- (2) Use the algorithm `SampPolys` from Lemma 4.1 to *uniformly sample* polynomials $g_0, \dots, g_L$ ($L = 1$ for BBS and $L = 2$ for BBS+) of degrees at most $\mathbf{q}L/(L+1)$ conditioning on $g_j(-e_i) = \mathbf{m}_i[j]g_0(-e_i)$ for all $i \in [\mathbf{q}]$ and $j \in [L]$. Note here that for $\mathcal{B}_2$, $\mathbf{m}_i = (m_i, s_i)$.

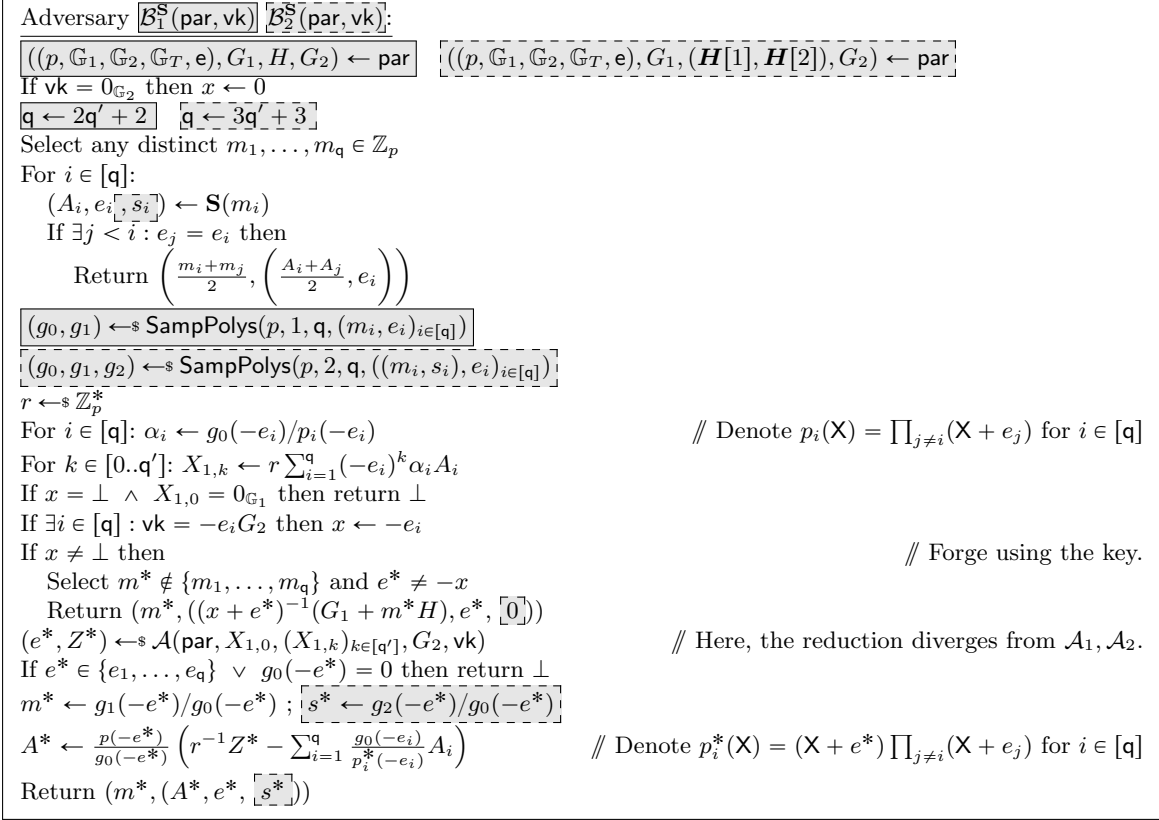


Fig. 5. Description of adversaries $\mathcal{B}_1$ and $\mathcal{B}_2$ in Theorem 5.1. The solid and dashed boxes denote the pseudocode exclusive to $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively. Both adversaries run a q' -SDH adversary $\mathcal{A}$ as a subroutine. The description of SampPolys is given in Figure 4.

- (3) By Lemma 4.2, we can linearly combine the signatures to compute a q' -SDH instance $X_{1,k} = x^k X_{1,0}$ for $k \in [0..q']$ with x denoting the secret key. Note that $X_{1,0}$ is also rerandomized with a scalar $r \in \mathbb{Z}_p^*$, so that it looks random and independent from (m_i, e_i, s_i) 's (we argue this formally later on).
- (4) Then, we run $\mathcal{A}$ on this instance to find (e^*, Z^*) such that $Z^* = \frac{1}{x+e^*} X_{1,0}$. The reduction aborts if $e^* \in \{e_1, \dots, e_q\}$.
- (5) Next, let $p_i^*(X) = (X + e^*) \prod_{j \neq i} (X + e_j)$ for $i \in [q]$. Also, compute A^* as a linear combination of Z^* and A_i such that

$$A^* = \frac{p(-e^*)}{g_0(-e^*)} \left(r^{-1} Z^* - \sum_{i=1}^q \frac{g_0(-e_i)}{p_i^*(-e_i)} A_i \right).$$

The final forgery is then (e^*, A^*) for the message $m^* = g_1(-e^*)/g_0(-e^*)$. (For BBS+, we also have that $s^* = g_2(-e^*)/g_0(-e^*)$.)

The running time of $\mathcal{B}_i$ for $i \in \{1, 2\}$ includes the running time of $\mathcal{A}$ and SampPolys , and the cost of computing $X_{1,k}$ for $k \in [0..q']$. Since each $X_{1,k}$ costs q exponentiation to compute, $\mathcal{B}_1$ and $\mathcal{B}_2$ run in time roughly $t_{\mathcal{A}} + O(q^2 T + q^3 T_p)$. We then analyze the success probabilities of $\mathcal{B}_1$ and $\mathcal{B}_2$.

Analysis of $\mathcal{B}_1$ and $\mathcal{B}_2$. First, without loss of generality, we assume that *the adversary $\mathcal{A}$ is deterministic*. We now consider the bad events that could occur during the run of $\mathcal{B}_1$ and $\mathcal{B}_2$.

Case 1: $x \in \{-e_1, \dots, -e_q\}$. In this case, we already find x and the reduction wins the game.

Case 2: CollE: $e_i = e_j$ for some $i \neq j \in [q]$. In this case, since $m_i \neq m_j$, we have a forgery for the message $(m_i + m_j)/2$ of the form $(e_i, A^* = (A_i + A_j)/2, (s_i + s_j)/2)$. Note that $\Pr[\text{CollE}] \leq q^2/2p$.

Case 3: ZeroX: none of the above occurs and $X_{1,0} = 0_{\mathbb{G}_1}$. In this case, we have (for the case of $\mathcal{B}_2$ which is more general)

$$0_{\mathbb{G}_1} = \frac{g_0(x)G_1 + g_1(x)\mathbf{H}[1] + g_2(x)\mathbf{H}[2]}{p(x)}.$$

Case 4: None of the above occurs and $-e^*G_2 = \text{vk}$. Similar to Case 1, we already find x and the reduction wins the game.

Case 5: BadE: none of the above occurs and $e^* \in \{e_1, \dots, e_q\}$. We analyze this case in the proof of Lemma 5.7.

Case 6: BadG: none of the above occurs and $g_0(-e^*) = 0$. We analyze this case in the proof of Lemma 5.8.

WHEN NO BAD EVENTS OCCUR. Finally, we consider when none of the events above occurs. In this case by CollE not occurring and Lemma 4.1, we have g_0, g_1 (and g_2 for BBS+) are such that $m_i g_0(-e_i) = g_1(-e_i)$ (and $s_i g_0(-e_i) = g_2(-e_i)$) for $i \in [q]$. Moreover, by Lemma 4.2, for $k \in [0..q']$

$$X_{1,k} = r \cdot x^k \left(\frac{g_0(x)G_1 + g_1(x)\mathbf{H}}{p(x)} \right).$$

Since ZeroX does not occur, $\mathcal{A}$ sees a random q' -SDH instance with the underlying discrete log related to x . This is because the reduction rerandomizes $X_{1,k}$ by multiplying them with a random scalar $r \leftarrow \mathbb{Z}_p^*$.

If $\mathcal{A}$ succeeds, we have $(e^*, Z^* = \frac{1}{x+e^*} X_{1,0})$. Next, if BadE and BadG do not occur, we have the following equation

$$\begin{aligned} r^{-1}Z^* &= \frac{g_0(x)G_1 + g_1(x)\mathbf{H}[1] + g_2(x)\mathbf{H}[2]}{(x+e^*)p(x)} \\ &= \frac{1}{(x+e^*)p(x)} \left(\frac{g_0(-e^*)}{p(-e^*)} p(x)G_1 + \sum_{i=1}^q \frac{g_0(-e_i)}{p_i^*(-e_i)} p_i^*(x)G_1 \right. \\ &\quad \left. + \frac{g_1(-e^*)}{p(-e^*)} p(x)\mathbf{H}[1] + \sum_{i=1}^q \frac{g_1(-e_i)}{p_i^*(-e_i)} p_i^*(x)\mathbf{H}[1] \right. \\ &\quad \left. + \frac{g_2(-e^*)}{p(-e^*)} p(x)\mathbf{H}[2] + \sum_{i=1}^q \frac{g_2(-e_i)}{p_i^*(-e_i)} p_i^*(x)\mathbf{H}[2] \right) \\ &= \frac{g_0(-e^*)G_1 + g_1(-e^*)\mathbf{H}[1] + g_2(-e^*)\mathbf{H}[2]}{(x+e^*)p(-e^*)} \\ &\quad + \sum_{i=1}^q \frac{g_0(-e_i)G_1 + g_1(-e_i)\mathbf{H}[1] + g_2(-e_i)\mathbf{H}[2]}{p_i^*(-e_i)(x+e_i)} \\ &= \frac{g_0(-e^*)G_1 + g_1(-e^*)\mathbf{H}[1] + g_2(-e^*)\mathbf{H}[2]}{(x+e^*)p(-e^*)} + \sum_{i=1}^q \frac{g_0(-e_i)}{p_i^*(-e_i)} A_i. \end{aligned}$$

The second equality follows from Lagrange basis polynomials with the basis being $p(X), p_1^*(X), \dots, p_q^*(X)$ and that the events $x \in \{-e^*, -e_1, \dots, -e_q\}$, BadE, CollE do not occur. (Note that this step is similar to the proof of Lemma 4.2.) The third equality follows from rearranging the terms. The fourth equality follows from how g_0, g_1, g_2 and the A_i 's are defined, as $m_i g_0(-e_i) = g_1(-e_i)$ and $s_i g_0(-e_i) = g_2(-e_i)$ for $i \in [q]$.

By the above equations with BadE and BadG not occurring, $p(-e^*) \neq 0$ and $g_0(-e^*) \neq 0$, so (e^*, A^*) is exactly a BBS signature of $(m^* = g_1(-e^*)/g_0(-e^*), s^* = g_2(-e^*)/g_0(-e^*))$.

ANALYSIS OF ZeroX, BadE, AND BadG. To conclude the proof, we analyze the probability of ZeroX, BadE, and BadG. We note first that when analyzing the events ZeroX and BadG, we have to analyze the probability that $g_0(x) = 0$ for some $x \in \mathbb{Z}_p$. We know that the coefficients of g_0, g_1 (and g_2 for $\mathcal{B}_2$), denoted $\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2$, are sampled (via SampPolys) such that

$$\mathbf{A}_{\text{BBS}}(\mathbf{g}_0, \mathbf{g}_1)^T = \mathbf{0}, \text{ and } \mathbf{A}_{\text{BBS}+}(\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2)^T,$$

where the matrices $\mathbf{A}_{\text{BBS}}$ and $\mathbf{A}_{\text{BBS}+}$ are defined as:

$$\mathbf{A}_{\text{BBS}} := \begin{pmatrix} -m_1 & \dots & -m_1 e_1^{q/2} & 1 & \dots & e_1^{q/2} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -m_q & \dots & -m_q e_q^{q/2} & 1 & \dots & e_q^{q/2} \end{pmatrix}, \quad (1)$$

$$\mathbf{A}_{\text{BBS}+} := \begin{pmatrix} -m_1 & \dots & -m_1 e_1^{2q/3} & 1 & \dots & e_1^{2q/3} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \mathbf{0} \\ -m_q & \dots & -m_q e_q^{2q/3} & 1 & \dots & e_q^{2q/3} \\ -s_1 & \dots & -s_1 e_1^{2q/3} & & & 1 & \dots & e_1^{2q/3} \\ \vdots & \ddots & \vdots & \mathbf{0} & & \vdots & \ddots & \vdots \\ -s_q & \dots & -s_q e_q^{2q/3} & & & 1 & \dots & e_q^{2q/3} \end{pmatrix}, \quad (2)$$

with $e_1, \dots, e_q$ and $s_1, \dots, s_q$ uniformly random in $\mathbb{Z}_p$. In particular, we state the following core lemma, proved in Section 5.3.⁶

Lemma 5.2. *Let $q \in \mathbb{N}$ and p be a prime. Let $m_1, \dots, m_q \in \mathbb{Z}_p$ be distinct, and $x \in \mathbb{Z}_p$ be an arbitrary scalar. Then,*

$$\begin{aligned} \Pr_{e_1, \dots, e_q \leftarrow \mathbb{Z}_p} [(1, x, \dots, x^{q/2}, \mathbf{0}) \in \text{RowSpan}(\mathbf{A}_{\text{BBS}})] &\leq \frac{q^2}{2p}, \\ \Pr_{e_1, \dots, e_q, s_1, \dots, s_q \leftarrow \mathbb{Z}_p} [(1, x, \dots, x^{2q/3}, \mathbf{0}) \in \text{RowSpan}(\mathbf{A}_{\text{BBS}+})] &\leq \frac{q^2 + 3q}{p}. \end{aligned}$$

Remark 5.3. The above lemma applies to any distinct $m_1, \dots, m_q \in \mathbb{Z}_p$ (not necessarily uniformly random), which makes our attack/reduction more general as it applies when given any q signatures with distinct messages.

The above lemma establishes the bound on the probability that the vector $(1, x, \dots, x^{q/2}, \mathbf{0})$ is in the row span of $\text{RowSpan}(\mathbf{A}_{\text{BBS}})$. In the case that this event occurs, it is always guaranteed that g_0 sampled from SampPolys will always satisfy $g_0(x) = 0$, as we can write $(1, x, \dots, x^{q/2}, \mathbf{0})$ as a linear combination of rows of $\mathbf{A}_{\text{BBS}}$, and we know that $\mathbf{A}_{\text{BBS}}(\mathbf{g}_0, \mathbf{g}_1)^T = \mathbf{0}$. The high-level idea of the proof in Section 5.3 is to argue that the determinant of the matrix obtained by stacking $\mathbf{A}_{\text{BBS}}$ with $(1, x, \dots, x^{q/2}, \mathbf{0})$ (and removing some columns) is zero with negligible probability, which implies that such matrix is full-row-rank with high-probability. However, because the form of $\mathbf{A}_{\text{BBS}}$ is somewhat contrived, this argument becomes quite involved.

The following lemma then claims that if such bad event does not occur, $g_0(x) = 0$ with negligible probability.

Lemma 5.4. *Let $q \in \mathbb{N}$ and p be a prime. Let $m_1, \dots, m_q \in \mathbb{Z}_p$ be distinct, and $x \in \mathbb{Z}_p$ be an arbitrary scalar. Also, let Good_A and Good_A^+ be the event defined over $e_1, \dots, e_q, s_1, \dots, s_q$ such that $(1, x, \dots, x^{q/2}, \mathbf{0}) \notin$*

⁶ The analysis of this lemma is the major distinction between the reductions to $\mathbf{q}'$ -DL and reductions to $\mathbf{q}'$ -SDH assumptions. Intuitively, the $\mathbf{q}'$ -DL reduction does not have to deal with the case due to the underlying adversary $\mathcal{A}$ outputting the discrete log x directly.

$\text{RowSpan}(\mathbf{A}_{\text{BBS}})$ and $(1, x, \dots, x^{2q/3}, \mathbf{0}) \notin \text{RowSpan}(\mathbf{A}_{\text{BBS}+})$, respectively. Then,

$$\begin{aligned} \Pr_{\substack{e_1, \dots, e_q \leftarrow \mathbb{Z}_p \\ (g_0, g_1) \leftarrow \text{SampPolys}(p, 1, q, (m_i, e_i)_{i \in [q]})}} [g_0(x) = 0 | \text{Good}_A] &= \frac{1}{p}, \\ \Pr_{\substack{e_1, \dots, e_q \leftarrow \mathbb{Z}_p \\ s_1, \dots, s_q \leftarrow \mathbb{Z}_p \\ (g_0, g_1, g_2) \leftarrow \text{SampPolys}(p, 2, q, ((m_i, s_i), e_i)_{i \in [q]})}} [g_0(x) = 0 | \text{Good}_A^+] &= \frac{1}{p}. \end{aligned}$$

Proof. First, let's fix $e_1, \dots, e_q$ and $s_1, \dots, s_q$ such that Good_A and Good_A^+ are true (note that the events are not determined by g_0, g_1, g_2). Then, for the case of BBS+

$$\begin{aligned} &\Pr_{(g_0, g_1, g_2) \leftarrow \text{SampPolys}(p, 2, q, ((m_i, s_i), e_i)_{i \in [q]})} [g_0(x) = 0] \\ &= \Pr_{g_0, g_1, g_2 \leftarrow \mathbb{Z}_p^{\leq 2q/3}[X]} [g_0(x) = 0 | \mathbf{A}_{\text{BBS}+}(g_0, g_1, g_2)^T = 0] \\ &= \frac{\Pr_{g_0, g_1, g_2} [g_0(x) = 0 \wedge \mathbf{A}_{\text{BBS}+}(g_0, g_1, g_2)^T = 0]}{\Pr_{g_0, g_1, g_2} [\mathbf{A}_{\text{BBS}+}(g_0, g_1, g_2)^T = 0]} \\ &= \Pr_{g_0, g_1, g_2} [g_0(x) = 0] \frac{\Pr_{g_0, g_1, g_2} [\mathbf{A}_{\text{BBS}+}(g_0, g_1, g_2)^T = 0]}{\Pr_{g_0, g_1, g_2} [\mathbf{A}_{\text{BBS}+}(g_0, g_1, g_2)^T = 0]} \\ &= \Pr_{g_0, g_1, g_2} [g_0(x) = 0] = \frac{1}{p}. \end{aligned}$$

The first equality follows from how SampPolys is defined. The second equality follows from the definition of conditional probability. The third equality follows from the fact that $\bar{x} = (1, x, \dots, x^{2q/3}, \mathbf{0}) \notin \text{RowSpan}(\mathbf{A}_{\text{BBS}+})$, i.e. $\bar{x}$ is linearly independent from rows of $\mathbf{A}_{\text{BBS}+}$, so the two events are independent. Thus,

$$\Pr_{\substack{e_1, \dots, e_q \leftarrow \mathbb{Z}_p \\ s_1, \dots, s_q \leftarrow \mathbb{Z}_p \\ (g_0, g_1, g_2) \leftarrow \text{SampPolys}(p, 2, q, ((m_i, s_i), e_i)_{i \in [q]})}} [g_0(x) = 0 | \text{Good}_A^+] = \frac{1}{p}.$$

Similarly, it follows that

$$\Pr_{\substack{e_1, \dots, e_q \leftarrow \mathbb{Z}_p \\ (g_0, g_1) \leftarrow \text{SampPolys}(p, 1, q, (m_i, e_i)_{i \in [q]})}} [g_0(x) = 0 | \text{Good}_A] = \frac{1}{p}. \quad \square$$

The following lemma gives an upperbound for the event ZeroX and is proven in Section 5.4. Note that we use the shorthand $\Pr_{\mathcal{B}_1}[\cdot]$ and $\Pr_{\mathcal{B}_2}[\cdot]$ to denote the probability of a specific event occurring during the run of $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively.

Lemma 5.5. For $\mathcal{B}_1$, $\Pr_{\mathcal{B}_1}[\text{ZeroX}] \leq \frac{q^2+4}{2p}$. For $\mathcal{B}_2$, $\Pr_{\mathcal{B}_2}[\text{ZeroX}] \leq \frac{q^2+3q+2}{p}$.

To show the bound on $\Pr[\text{BadE}]$ and $\Pr[\text{BadG}]$, we first prove the following lemma showing the statistical distance (which we denote SD and is defined as $\text{SD}(\mathcal{D}_0, \mathcal{D}_1) = \sum_{\tau} \max\{0, \Pr_{T_0 \leftarrow \mathcal{D}_0}[T_0 = \tau] - \Pr_{T_1 \leftarrow \mathcal{D}_1}[T_1 = \tau]\}$) between the two distributions of the following values for fixed distinct messages $m_1, \dots, m_q \in \mathbb{Z}_p$,

$$(G_1, G_2, \mathbf{H}[1], \mathbf{H}[2], x, (m_i, e_i, s_i)_{i \in [q]}, (X_{1,i})_{i \in [0..q]}, X_{2,i}, g_0, g_1, g_2),$$

where one is generated as $\mathcal{B}_1, \mathcal{B}_2$ would, and another is generated as in Figure 6. Note that the second distribution is sampled with the inputs $(X_{1,i})_{i \in [q]}, X_{2,i}$ to $\mathcal{A}$ being independent from e_i, s_i, g_0, g_1, g_2 .

Lemma 5.6. Let $m_1, \dots, m_q \in \mathbb{Z}_p$ be distinct scalars, and $\mathcal{D}_{0,\text{BBS},\mathbf{m}}, \mathcal{D}_{1,\text{BBS},\mathbf{m}}, \mathcal{D}_{0,\text{BBS}+,\mathbf{m}}$, and $\mathcal{D}_{1,\text{BBS}+,\mathbf{m}}$ (note that we will drop the subscript $\mathbf{m}$ when clear from the context) be the distributions from which are sampled as described in Figure 6. Then,

$$\text{SD}(\mathcal{D}_{0,\text{BBS},\mathbf{m}}, \mathcal{D}_{1,\text{BBS},\mathbf{m}}) \leq \frac{q^2 + q + 2}{p}, \text{ and } \text{SD}(\mathcal{D}_{0,\text{BBS}+,\mathbf{m}}, \mathcal{D}_{1,\text{BBS}+,\mathbf{m}}) \leq \frac{3q^2 + 8q + 4}{2p}.$$

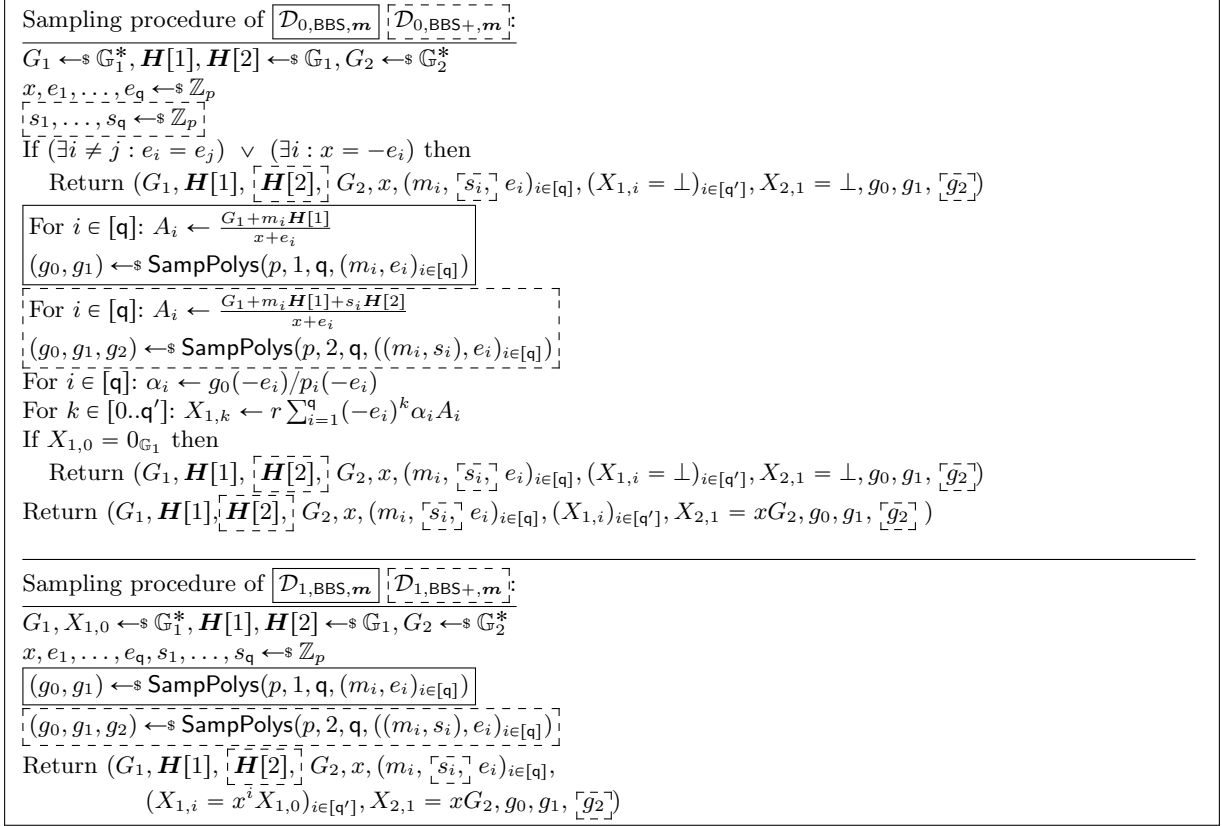


Fig. 6. Sampling procedures for the distributions in Lemma 5.6 where $\mathcal{D}_{b,\text{BBS}}$ contains everything except dashed boxes, while $\mathcal{D}_{b,\text{BBS}^+}$ contains everything except solid boxes.

Proof (of Lemma 5.6). The proofs for BBS and BBS+ follow the same structure, so we show that for BBS+ for more generality. Namely, we consider transcripts of the form

$$\hat{T} = (\hat{G}_1, \hat{\mathbf{H}}[1], \hat{\mathbf{H}}[2], \hat{G}_2, \hat{x}, (\hat{m}_i, \hat{e}_i, \hat{s}_i)_{i \in [q]}, (\hat{X}_{1,i})_{i \in [q]}, \hat{X}_{2,1}, \hat{g}_0, \hat{g}_1, \hat{g}_2),$$

such that the following conditions are satisfied

1. $\hat{X}_{1,i} = \hat{x}^i \hat{X}_{1,0}$ for $i \in [q]$
2. $\hat{x} \notin \{-\hat{e}_1, \dots, -\hat{e}_q\}$
3. $\hat{e}_i \neq \hat{e}_j$ for all $i \neq j \in [q]$
4. $\frac{\hat{g}_0(x)G_1 + \hat{g}_1(x)\mathbf{H}[1] + \hat{g}_2(x)\mathbf{H}[2]}{p(x)} \neq 0_{\mathbb{G}_1}$

Denote

$$\rho(\hat{T}) := \Pr_{(g_0, g_1, g_2) \leftarrow \text{SampPolys}(p, 2, q, (\hat{m}_i, \hat{s}_i, \hat{e}_i)_{i \in [q]})} [\forall j \in [0..2] : g_j = \hat{g}_j]$$

as the probability that SampPolys sample $\hat{g}_0, \hat{g}_1, \hat{g}_2$ when the inputs m_i, s_i, e_i are fixed as in $\hat{T}$.

Now, the probability that we get this transcript $\hat{T}$ sampling from $\mathcal{D}_{1,\text{BBS}^+}$ is exactly $\frac{1}{(p-1)^3 p^{2q+3}} \rho(\hat{T})$ as each $G_1, \mathbf{H}[1], \mathbf{H}[2], G_2, x, (e_i, s_i)_{i \in [q]}, X_{1,0}$ are uniformly random in their respective domain and g_0, g_1, g_2 are sampled from the algorithm SampPolys .

For $\mathcal{D}_{0,\text{BBS}+}$, $X_{1,0}$ is exactly $r \cdot \frac{g_0(x)G+g_1(x)\mathbf{H}[1]+g_2(x)\mathbf{H}[2]}{p(x)}$ for $r \leftarrow \mathbb{Z}_p^*$ and $G_1, \mathbf{H}[1], \mathbf{H}[2], G_2, x, (e_i, s_i)_{i \in [q]}$ are uniformly random in their respective domains, and g_0, g_1, g_2 are sampled from SampPolys . Hence, the probability that the sampled transcript is $\hat{T}$ is $\frac{1}{(p-1)^3 p^{2q+3}} \rho(\hat{T})$.

Moreover, since in $\mathcal{D}_{0,\text{BBS}}$, the events $x \in \{-e_1, \dots, -e_q\}$, $e_i = e_j$ for some $i \neq j$, and ZeroX occurs with probability at most $q/p, q^2/2p, (q^2 + 4)/2p$. With the fact that the probabilities are equal for transcripts that does not trigger such events, the statistical distance is at most $(q^2 + q + 2)/p$. For $\mathcal{D}_{0,\text{BBS}+}$, the statistical distance is at most $(3q^2 + 8q + 4)/2p$. $\square$

Next, we define BadE' as the event defined on transcripts sampled from $\mathcal{D}_{1,\text{BBS}}$ or $\mathcal{D}_{1,\text{BBS}+}$ such that after sampling the transcript T , we run the adversary $\mathcal{A}$ on input $((X_{1,i})_{i \in [0..q]}, G_2, X_{2,1})$ and the output (e^*, Z^*) of $\mathcal{A}$ is such that $e^* \in \{e_1, \dots, e_q\}$. Similarly, we also define the event BadG' on $\mathcal{D}_{1,\text{BBS}}$ or $\mathcal{D}_{1,\text{BBS}+}$ such that after sampling the transcript, running $\mathcal{A}$, and sampling (g_0, g_1, g_2) from SampPolys on input $((m_i, s_i), e_i)_{i \in [q]}$, $g_0(e^*) \neq 0$. Note that these events are analogous to the events BadE and BadG defined with respect to $\mathcal{B}_1$ and $\mathcal{B}_2$. The following lemmas establish a bound on the probability of these bad events. Also, we denote the shorthand $\Pr_{\mathcal{D}}[\cdot]$ as the probability that an event occurs with respect to the random variables sampled from the distribution $\mathcal{D}$.

Lemma 5.7. $\Pr_{\mathcal{D}_{1,\text{BBS}}}[\text{BadE}'] = \Pr_{\mathcal{D}_{1,\text{BBS}+}}[\text{BadE}'] \leq \frac{q}{p}$.

Proof. Note that the distribution of inputs to $\mathcal{A}$ is independent of $e_1, \dots, e_q$. Thus, fixing an arbitrary $e^* \in \mathbb{Z}_p$ and with e_i 's uniformly random in $\mathbb{Z}_p$, the lemma follows by the union bound over $i \in [q]$. $\square$

Lemma 5.8. $\Pr_{\mathcal{D}_{1,\text{BBS}}}[\text{BadG}'] \leq \frac{q^2+2}{2p}$ and $\Pr_{\mathcal{D}_{1,\text{BBS}+}}[\text{BadG}'] \leq \frac{q^2+3q+1}{p}$.

We refer to Section 5.5 for the proof.

Note that by Lemmas 5.6 to 5.8,

$$\begin{aligned} & \Pr_{\mathcal{B}_1}[\text{CollE} \vee \text{ZeroX} \vee \text{BadE} \vee \text{BadG}] \\ & \leq \Pr[\text{CollE}] + \Pr_{\mathcal{B}_1}[\text{ZeroX}] + \Pr_{\mathcal{D}_{1,\text{BBS}}}[\text{BadE}' \vee \text{BadG}'] + \text{SD}(\mathcal{D}_{0,\text{BBS}}, \mathcal{D}_{1,\text{BBS}}) \\ & \leq \frac{q^2}{2p} + \frac{q^2 + 4}{2p} + \frac{q^2 + 2}{2p} + \frac{q}{p} + \frac{q^2 + q + 2}{p} = \frac{5q^2 + 4q + 10}{2p}, \end{aligned}$$

The first and second inequality follows from the union bound and property of statistical distance. Similarly,

$$\begin{aligned} & \Pr_{\mathcal{B}_2}[\text{CollE} \vee \text{ZeroX} \vee \text{BadE} \vee \text{BadG}] \\ & \leq \frac{q^2}{2p} + \frac{q^2 + 3q + 2}{p} + \frac{q}{p} + \frac{q^2 + 3q + 1}{p} + \frac{3q^2 + 8q + 4}{2p} \\ & \leq \frac{4q^2 + 11q + 5}{p}. \end{aligned}$$

Hence, we conclude that

$$\begin{aligned} \text{Adv}_{\text{BBS}}^{\text{uf}}(\mathcal{B}_1, \lambda) & \geq \text{Adv}_{\text{GGen}}^{q'\text{-sdh}}(\mathcal{A}, \lambda) - \frac{5q^2 + 4q + 10}{2p}, \\ \text{Adv}_{\text{BBS}+}^{\text{uf}}(\mathcal{B}_2, \lambda) & \geq \text{Adv}_{\text{GGen}}^{q'\text{-sdh}}(\mathcal{A}, \lambda) - \frac{4q^2 + 11q + 5}{p}. \square \end{aligned}$$

5.3 Proof of Lemma 5.2

The main idea of the proof is to consider the extended matrix which appends $\mathbf{x}$ to the matrix $\mathbf{A}_{\text{BBS}}$ (and $\mathbf{A}_{\text{BBS}+}$ in the other case). Then, we will show that the matrix is full-row-rank with high probability by considering the determinant of a square matrix obtained by truncating some columns of the extended matrix. We will show that this determinant can be written as a non-zero polynomial in the variables $\mathbf{E}_1, \dots, \mathbf{E}_q, \mathbf{S}_1, \dots, \mathbf{S}_q$ and apply Schwartz-Zippel lemma. We will show this for $\mathbf{A}_{\text{BBS}}$ and later on for $\mathbf{A}_{\text{BBS}+}$.

PROBABILITY BOUND FOR $\mathbf{A}_{\text{BBS}}$: Consider the following matrix extended from $\mathbf{A}_{\text{BBS}}$ with the $q/2 + 2$ -th column truncated, which when replacing e_i with a formal variable E_i is

$$\mathbf{A}' := \begin{pmatrix} -m_1 \dots -m_1 E_1^{q/2} & E_1 & \dots & E_1^{q/2} \\ \vdots & \ddots & \vdots & \vdots \\ -m_q \dots -m_q E_q^{q/2} & E_q & \dots & E_q^{q/2} \\ 1 & \dots & x^{q/2} & 0 \end{pmatrix}.$$

If $\det \mathbf{A}' \neq 0$, applying Schwartz-Zippel lemma with $\det \mathbf{A}'$ having degree at most $(q/2 + 1)q/2 \leq q^2/2$, we have the bound for $\mathbf{A}_{\text{BBS}}$. To show that $\det \mathbf{A}' \neq 0$, we will show that there exists a non-zero monomial in the expression of $\det \mathbf{A}'$; in particular, we will show that the coefficient of the monomial $\mathbf{M} = \prod_{i=1}^{q/2} E_i E_{i+q/2}^i$ is non-zero. To do so, we will be considering the determinant formula of $\mathbf{A}'$ by Laplace expansion: for any $j \in [q + 1]$

$$\det \mathbf{A}' = \sum_{i=1}^{q+1} (-1)^{i+j} A'_{j,i} \det \mathbf{B}^{(j,i)},$$

where $\mathbf{B}^{(j,i)}$ is the matrix $\mathbf{A}'$ with row j and column i removed. Note that fixing $j = q + 1$ (i.e., the last row) the monomial $\mathbf{M}$ only appears in $\det \mathbf{B}^{(q+1,1)}$ as $\det \mathbf{B}^{(q+1,i)}$ for $i > 1$ contains terms with lower degree. Moreover, its coefficient is $(-1)^{i+j} c$ where c is its coefficient in $\det \mathbf{B}^{(q+1,1)}$, meaning we only need to show that $c \neq 0$.

To do this, we consider the Leibniz formula of determinant and write

$$\det \mathbf{B}^{(q+1,1)} = \sum_{\rho} \text{sgn}(\rho) \prod_i B_{i,\rho(i)}^{(q+1,1)},$$

where the summation is over permutations ρ of $[q]$ and $\text{sgn}(\rho)$ denotes the parity of the permutation which is defined by (a) $\text{sgn}(\text{id})$ where id is the identity map and (b) $\text{sgn}(\rho) = -\text{sgn}(\rho')$ for ρ, ρ' such that there exists $i \neq j$ where $\rho(i) = \rho'(j), \rho(j) = \rho'(i)$ and $\rho(k) = \rho'(k)$ for all $k \in [q] \setminus \{i, j\}$. Also, to illustrate the argument, we write the matrix

$$\mathbf{B}^{(q+1,1)} := \begin{pmatrix} -m_1 E_1 \dots -m_1 E_1^{q/2} & E_1 & \dots & E_1^{q/2} \\ \vdots & \ddots & \vdots & \vdots \\ -m_q E_q \dots -m_q E_q^{q/2} & E_q & \dots & E_q^{q/2} \end{pmatrix}.$$

Claim. Let $\rho : [q] \rightarrow [q]$ be a permutation. $\prod_i B_{i,\rho(i)}^{(q+1,1)} = c_{\rho} \mathbf{M}$ for some constant $c_{\rho} \in \mathbb{Z}_p$ if and only if

$$\rho(i) \in \begin{cases} \{i, i + q/2\} & \text{if } 1 \leq i \leq q/2 \\ \{i, i - q/2\} & \text{otherwise} \end{cases}. \quad (3)$$

This can be easily observed from the structure of $\mathbf{B}^{(q+1,1)}$. Now, with this claim, we can write the coefficient of $\mathbf{M}$ as a sum of $2^{q/2}$ products, each corresponding to a permutation $\rho_{b_1, \dots, b_{q/2}}$ depending on bits $b_1, \dots, b_{q/2} \in \{0, 1\}$ such that $\rho_{b_1, \dots, b_{q/2}}(i) = i + b_i q/2$ for $i \in [q/2]$. Also, by the definition of sgn , $\text{sgn}(\rho_{b_1, \dots, b_{q/2}}) = (-1)^{\sum b_i}$. Now, the coefficient is

$$\begin{aligned} c_{\mathbf{M}} &:= (-1)^{q/2} \sum_{b_1, \dots, b_{q/2}} \text{sgn}(\rho_{b_1, \dots, b_{q/2}}) \prod_{i=1}^{q/2} m_{\rho_{b_1, \dots, b_{q/2}}(i)} \\ &= (-1)^{q/2} \sum_{b_1, \dots, b_{q/2}} (-1)^{\sum_{i=1}^{q/2} b_i} \prod_{i=1}^{q/2} m_{i + b_i \cdot q/2}. \end{aligned}$$

Notice that $c_{\mathbf{M}} = (-1)^{q/2} \prod_{i=1}^{q/2} (m_i - m_{i+q/2})$, which is non-zero because $m_i \neq m_j$ for all $i \neq j$. Thus, concluding the proof for $\mathbf{A}_{\text{BBS}}$

PROBABILITY BOUND FOR $\mathbf{A}_{\text{BBS}+}$: We take similar approach to the above argument, but now we have that the scalars $s_1, \dots, s_q$ are uniformly random. In particular, we consider the determinant of the following matrix $\mathbf{A}''$ which is $\mathbf{A}_{\text{BBS}+}$ appended with a row vector $\bar{x} = (1, x, \dots, x^{2q/3}, \mathbf{0})$ and truncating the degree 0 columns that is not the first one.

$$\mathbf{A}'' := \begin{pmatrix} -m_1 \dots -m_1 E_1^{2q/3} & E_1 \dots E_1^{2q/3} & & & \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \mathbf{0} \\ -m_q \dots -m_q E_q^{2q/3} & E_q \dots E_q^{2q/3} & & & \\ S_1 \dots S_1 E_1^{2q/3} & & E_1 \dots E_1^{2q/3} & & \\ \vdots & \ddots & \vdots & \mathbf{0} & \vdots & \ddots & \vdots \\ S_q \dots S_q E_q^{2q/3} & & E_q \dots E_q^{2q/3} & & \\ 1 \dots x^{2q/3} & 0 \dots 0 & 0 \dots 0 & & \end{pmatrix}.$$

In particular, we want to show that the coefficient of the monomial $M' = \prod_{i=1}^{q/3} S_{i+2q/3} E_i^{2i} E_{i+q/3}^{2q/3+2i}$ is non-zero (for example, for $q = 3$, this term is $S_3 E_1^2 E_2^3 E_3^4$). Following a similar argument, we first show that the term M' can only occur in $\det \mathbf{B}^{(2q+1,1)}$ contained in the sum $\det \mathbf{A}''$ where $\mathbf{B}^{(2q+1,1)}$ being the matrix obtained by removing from $\mathbf{A}''$ the row $2q+1$ and column 1. This is due to the degree of the E_i of such term being

$$\sum_{i=1}^{q/3} (6i + q) = q^2/3 + 2q(q+3)/3 = q^2 + 2q.$$

However, for any other $\mathbf{B}^{(2q+1,j)}$ for $j > 1$, we have that the maximum degree (excluding the degree of S_i) for monomials containing E_i is less than $q^2 + 2q$. Thus, M' can only occur in $\det \mathbf{B}^{(2q+1,1)}$.

We again consider the Leibniz formula of determinant and write

$$\det \mathbf{B}^{(2q+1,1)} = \sum_{\rho} \text{sgn}(\rho) \prod_i B_{i, \rho(i)}^{(2q+1,1)}$$

Now, we show an analogous claim to the case above.

Claim. Let $\rho : [2q] \rightarrow [2q]$ be a permutation. $\prod_i B_{i, \rho(i)}^{(2q+1,1)} = c_{\rho} M'$ for some constant $c_{\rho} \in \mathbb{Z}_p$ if and only if

$$\rho(i) \in \begin{cases} \{i, i + 2q/3\} & \text{for } 1 \leq i \leq q/3 \\ \{i - q/3, i + q/3\} & \text{for } q/3 < i \leq 2q/3 \\ \{i + q/3\} & \text{for } 2q/3 < i \leq q \\ \{i + q/3\} & \text{for } q < i \leq 4q/3 \\ \{i + q/3\} & \text{for } 4q/3 < i \leq 5q/3 \\ \{i - 4q/3\} & \text{for } 5q/3 < i \leq 2q \end{cases}. \quad (4)$$

Proof (of Claim). ($\Leftarrow$) The ‘only if’ direction can be proved by counting the degree of S_i, E_i defined by the assignments in ρ .

($\Rightarrow$) For the ‘if’ direction, we prove the claim by considering the degree of S_i and E_i . Note that the positions defined by ρ are visualized in Figure 7

First, consider for $2q/3 < i \leq q$ that M' contains S_i and E_i has degree $2q/3 + 2(i - 2q/3)$. Containing S_i implies that $q/3 + 1 \leq \rho(i + q) \leq 2q/3$ (since those are the only columns containing S_i). Now, we consider the degree of E_q which is $4q/3$ (which is the maximum degree for a variable one could have in the determinant). By the previous observation, we have that $\rho(2q) = 2q/3$ and $\rho(q) = 4q/3$. Inductively, since columns $2q/3$ and $4q/3$ are mapped to by ρ , we can argue that for $2q/3 < i \leq q$

$$\rho(i) = i + q/3 \wedge \rho(i + q) = i - q/3, \quad (5)$$

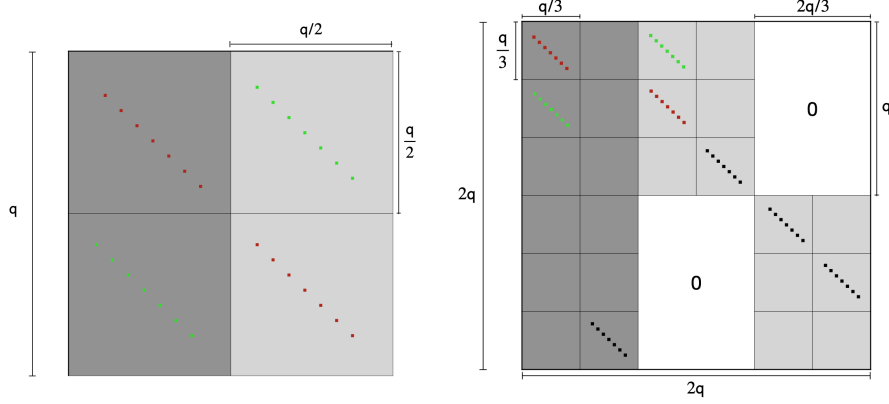


Fig. 7. Visualization of the possible permutations ρ as defined in Equations (3) and (4) (on the left and right, respectively). The red and green dots refer to the two possible column assignments of the designated rows.

proving the third and last cases.

Next, consider for $1 \leq i \leq 2q/3$ that M' does not contain S_i , meaning

$$4q/3 < \rho(i + q) \leq 2q. \quad (6)$$

Now, we again consider the degree of E_i for $1 \leq i \leq q/3$ which is $2i$. By similarly arguing that this is only achievable by selecting the minimum degree column available, we have that $\rho(i)$ and $\rho(i + q)$ maps to a column of degree i —see first that the degree of E_1 is 2 and the minimum degree possible is by selecting two columns of degree 1. With the given derived constraints $1 \leq i \leq q/3$

$$\rho(i) \in \{i, i + 2q/3\} \wedge \rho(i + q) = i + 4q/3. \quad (7)$$

proving the first and fourth cases.

Finally, we consider the degree of E_i for $q/3 < i \leq 2q/3$ which is $2i - q/3$. With Equations (6) and (7), we have

$$5q/3 < \rho(i + q) \leq 2q. \quad (8)$$

Then, we employ similar argument observing first that the minimum degree of $E_{q/3+1}$ is $q/3 + 2$ which is possible only if columns of degree 1 and $q/3 + 1$ are selected (which are the only ones left). This implies that $\rho(q/3 + 1) = 1$ or $2q/3 + 1$ and $\rho(4q/3 + 1) = 5q/3 + 1$. Inductively, we conclude that for $q/3 < i \leq 2q/3$

$$\rho(i) \in \{i - q/3, i + q/3\} \wedge \rho(i + q) = i + 4q/3,$$

proving the claim. $\square$

According to the above claim, for $1 \leq i \leq q/3$, $\rho(i) = i$ if and only if $\rho(i + q/3) = i + 2q/3$. Also, let ρ_0 be a permutation satisfying the conditions in the claim with $\rho_0(i) = i$ for all $1 \leq i \leq q/3$. Also, define each permutation ρ using bits $b_1, \dots, b_{q/3} \in \{0, 1\}$ such that $\rho(i) = \rho_0(i)$ iff $b_i = 0$ for all $i \in [q/3]$. Then, the coefficient $c_{M'}$ can be written as

$$\begin{aligned} c_{M'} &:= (-1)^{q/3} \sum_{b_1, \dots, b_{q/3}} \text{sgn}(\rho_{b_1, \dots, b_{q/3}}) \prod_{i=1}^{q/3} m_{\rho_{b_1, \dots, b_{q/3}}(i)} \\ &= (-1)^{q/3} \text{sgn}(\rho_0) \sum_{b_1, \dots, b_{q/3}} (-1)^{\sum_{i=1}^{q/3} b_i} \prod_{i=1}^{q/3} m_{i + b_i \cdot q/3}, \end{aligned}$$

We can see that $\sum_{b_1, \dots, b_{q/3}} (-1)^{\sum_{i=1}^{q/3} b_i} \prod_{i=1}^{q/3} m_{i+b_i \cdot q/3} = \prod_{i=1}^{q/3} (m_i - m_{i+q/3})$ which is non-zero by the distinctness of m_i 's. Thus, we have a non-zero monomial in $\det \mathbf{A}''$.

Finally, we can conclude using Schwartz-Zippel lemma that the probability we want to bound is at most $\frac{q^2+3q}{p}$ (sum of degree of E_i is $q^2 + 2q$ and sum of degree of S_i is at most q). $\square$

5.4 Proof of Lemma 5.5

Recall that the event **ZeroX** corresponds to the event that

$$X_{1,0} = \frac{g_0(x)G_1 + g_1(x)\mathbf{H}[1] + g_2(x)\mathbf{H}[2]}{p(x)} = 0_{\mathbb{G}_1}.$$

In particular, we have two subcases to consider:

- $g_0(x) \neq 0$ or $g_1(x) \neq 0$ or $g_2(x) \neq 0$. In this case, we know $G_1, \mathbf{H}[1], \mathbf{H}[2]$ are independent from $e_1, \dots, e_q, s_1, \dots, s_q, g_0, g_1, g_2$ and they are uniformly random in $\mathbb{G}_1^*$ and $\mathbb{G}_1$, respectively. Therefore, the probability that $X_{1,0} = 0_{\mathbb{G}_1}$ in this case is at most $1/p$.
- $g_0(x) = g_1(x) = g_2(x) = 0$. In this case, we rely on Lemmas 5.2 and 5.4, i.e.,

$$\begin{aligned} & \Pr[g_0(x) = g_1(x) = g_2(x) = 0] \\ & \leq \Pr[g_0(x) = 0] \\ & \leq \Pr[g_0(x) = 0 | \text{Good}_A^+] \Pr[\text{Good}_A^+] + \Pr[\neg \text{Good}_A^+] \\ & \leq \frac{1}{p} + \frac{q^2 + 3q}{p}, \end{aligned}$$

The first inequality is simply by implication. The second one is by law of total probability and $\Pr[g_0(x) = 0 | \neg \text{Good}_A^+] \leq 1$. The last inequality follow from the referred lemmas. For the case of $\mathcal{B}_1$, we have the bound of $(q^2 + 2)/2p$.

Combining the two cases, we have $\Pr_{\mathcal{B}_1}[\text{ZeroX}] \leq (q^2 + 4)/2p$, and $\Pr_{\mathcal{B}_2}[\text{ZeroX}] \leq (q^2 + 3q + 2)/p$. $\square$

5.5 Proof of Lemma 5.8

First, we remark again that we are working with the distribution $\mathcal{D}_{1,\text{BBS}}$ and $\mathcal{D}_{1,\text{BBS}+}$. Since the input to $\mathcal{A}$, namely $((X_{1,i})_{i \in [0..q]}, G_2, X_{2,1})$, is independent of the sampled polynomial g_0 , we w.l.o.g. consider output e^* of $\mathcal{A}$ to be fixed. Now, the bad event $g_0(e^*) = 0$ can occurs if (a) the vector $(1, e^*, \dots, e^{*q/2}, \mathbf{0})$ or $(1, e^*, \dots, e^{*2q/3}, \mathbf{0})$ is in the row span of $\mathbf{A}_{\text{BBS}}$ or $\mathbf{A}_{\text{BBS}+}$ (as defined in Equations (1) and (2)), respectively, or (b) the prior does not occur, but we sample g_0 which has e^* as a root. Due to Lemmas 5.2 and 5.4, the former occurs with probability at most $\frac{q^2}{2p}$ and $\frac{q^2+3q}{p}$, respectively for $\mathbf{A}_{\text{BBS}}$ and $\mathbf{A}_{\text{BBS}+}$. The latter occurs with probability $1/p$, implying the bound in the lemma. $\square$

6 Discussion

In this section, we explore whether other scenarios may be applicable to our attack and further implications of our results.

Table 1. Estimated security level for BBS/BBS+ implementation on popular curves against our attack from Corollary 4.5. We assume the number of signatures on distinct messages obtained is roughly $q \approx 2^{40}$. (Note that in practice, one might even assume a larger bound on q .) $\dagger$: The parentheses denotes the value of suitable factor.

Elliptic Curve	Bit-size of suitable factors † of ($p - 1$) or ($p + 1$)	Estimated security level
BLS12-381	37 ($2^{32} \times 3 \times 11$)	110
BN254	36 ($2 \times 3^2 \times 13 \times$ $29 \times 67 \times 229 \times 311$)	110
secp256k1	35 ($2 \times 45751 \times 509879$)	112
secp256r1	38 ($2^3 \times 3 \times 71 \times 131 \times 373 \times 3407$)	109

IMPLICATIONS ON PROOFS OF KNOWLEDGE FOR BBS. We note that our attack *does not* apply (and cannot be extended) to the setting where adversaries only have access to the zero-knowledge proof of knowledge (ZKPoK) of BBS signatures (see e.g., protocols given in [TZ23]). More concretely, one may consider a security notion where adversaries (1) have oracle access to some oracle Prove which generates a fresh BBS signature on message of the adversary’s choice and return a ZKPoK of such signature and (2) try to give a valid forgery on a message not queried before. Immediately, one can see that such security is secure based on unforgeability of BBS *under no message attack* (i.e., adversary does not make any signing query), which is tightly implied by the 1-SDH assumption [TZ23]. This is due to the existence of zero-knowledge simulator of the ZKPoK. Hence, this implies generic group lower-bound of at least $O(\sqrt{p})$ group operations to forge a signature—excluding better generic attacks than breaking discrete logarithm. However, such security model is insufficient in practice, since adversaries can ask for credentials/signatures of their own attributes.

RELATIONSHIP TO UBER ASSUMPTIONS. Uber assumptions family [Boy08, BFL20] generalizes a family of q -type assumptions in prime-order (bilinear) groups. Prior works [JR15, GG17, BFL20] showed implications between assumptions within the uber family. Bauer, Fuchsbauer, and Loss [BFL20] recently showed that *in the AGM*, the q -DL family of assumptions implies all assumption in the Uber assumption family. In the standard model, Joux and Rojatz [JR15] showed implications between variants of the CDH assumption, and Ghadafi and Groth [GG17] classified all non-interactive uber assumptions based on q -Generalized Diffie-Hellman and q -Simple Fractional assumptions. Most of these showed that q -DL (or similar assumptions) implies all (or a broad class of) uber assumptions.

BBS and BBS+ security games can also be viewed as an uber assumption with the following (simplified) format:

- The adversary is given multiple generators $G_1, \mathbf{H}, G_2, X_{2,1}$ —denoting $\mathbf{h}$ and x as the discrete logarithms $\text{DL}_{G_1} \mathbf{H}$ and $\text{DL}_{G_2} X_{2,1}$, respectively.
- The adversary has access to an oracle which on input a rational function $f(\mathbf{H}_1, \dots, \mathbf{H}_\ell, \mathbf{X})$ of the form

$$f(\mathbf{H}_1, \dots, \mathbf{H}_\ell, \mathbf{X}) := \frac{1 + \sum_{i=1}^{\ell} \mathbf{m}[i] \mathbf{H}_i}{\mathbf{X} + e},$$

for some $\mathbf{m} \in \mathbb{Z}_p^\ell$ and uniformly random $e \leftarrow \mathbb{Z}_p$, returns a group element $A = f(\mathbf{h}, x)G_1$.

- The adversaries goal is to find a rational function f^* in the same format as the queries, but distinct, along with a group element A^* such that $A^* = f^*(\mathbf{h}, x)G_1$.

Our results can be casted as showing that this class of uber assumptions (with input group elements formulated as BBS/BBS+ signatures) implies the q -DL and q -SDH assumptions *in the standard model*.

IMPLICATION TO SECURITY LEVEL OF BBS. We consider the resulting security levels of BBS/BBS+ signatures with respect to implementations using concrete pairing-friendly elliptic curves. In particular, we consider the BLS12-381 [Bow17, BLS03] and BN254 [BN06] curves. The former is the one mentioned in the standardization process of BBS [LKWL25]. The latter was used in ZCash [BCG+14]. Additionally, we consider the security level of the algebraic MAC version of BBS [BBDT16, Orr24] when implemented on

pairing-free curves. These are `secp256k1` (used in Bitcoin) and the NIST curve `secp256r1` (used in the test implementation of BBS# [DDKT25]). For pairing-free curve `curve25519` [Ber06] and $q = 2^{40}$, the group order p has at most 8-bit divisor for $p - 1$ and 12-bit divisor for $p + 1$; however, if we consider $q = 2^{60}$, there is a 57-bit divisor of $p + 1$. In Table 1, we give the estimated security level with respect to our attacks. The group order p for each curve is detailed below in hexa-decimal:

- BLS12-381: 0x73EDA753299D7D483339D80809A1D80553BDA402FFFE5BFEFFFFFFF00000001
- BN254: 0x30644E72E131A029B85045B68181585D97816A916871CA8D3C208C16D87CFD47
- secp256k1: 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141
- secp256r1: 0xFFFFFFFFF00000000FFFFFFFFFFFFFFFFFBCE6FAADA7179E84F3B9CAC2FC632551

Acknowledgements

The authors wish to thank Dennis Hofheinz and Kenneth Paterson for discussions at the earlier stage of the work. This work was partially supported by NSF grants CNS-2026774, CNS-2154174, CNS-2426905, a gift from Microsoft, and a Stellar Development Foundation Academic Research Award.

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